

# Microsoft Azure Fundamentals (AZ-900) — Section 4 Lab

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## Azure Compute Options: VMs, App Service, Functions & Cost Analysis

**Certification Track:** Microsoft Azure Fundamentals (AZ-900) **Platform:** Microsoft Azure (Free Account)  
**Estimated Duration:** 25 minutes **Tools Used:** Azure Portal · Azure Pricing Calculator · Resource Groups  
· Virtual Machines · App Service · Azure Functions · Azure Advisor · Cost Management **Completed by:**  
**Oluwaseun Osunsola** [[LinkedIn](#)]

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### Lab Overview

In this hands-on lab, I acted as IT consultant for **CloudFirst Retail**, tasked with evaluating Azure compute options before committing to any infrastructure. The goal was to explore VM configuration in depth, compare VM sizing families and pricing across development vs. production workloads, understand the IaaS → PaaS → Serverless spectrum, and navigate cost optimization tools — all without deploying live resources or incurring charges.

This is the **planning and analysis phase** every cloud architect completes before production deployment.

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### Learning Objectives

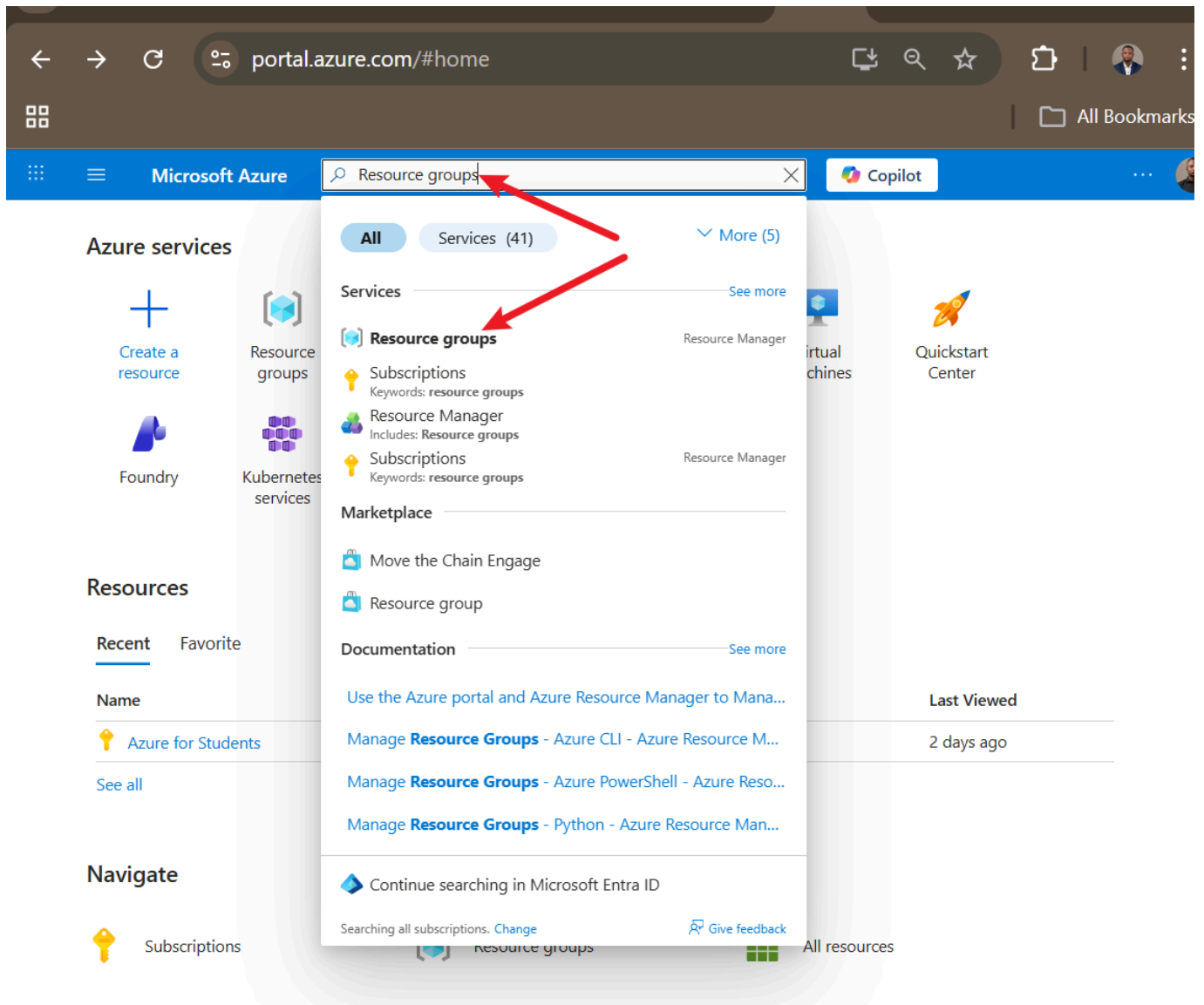
- Create a tagged Resource Group for a development environment
  - Explore full VM configuration options (Basics, Disks, Networking, Management, Monitoring, Tags) without deploying
  - Use the Azure Pricing Calculator to compare B, D, and E-series VM costs and savings plans
  - Navigate Azure Advisor's five optimization pillars
  - Explore App Service (PaaS) and Azure Functions (Serverless) creation flows and pricing tiers
  - Navigate Cost Management + Billing and apply tag-based cost filters
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### Step-by-Step Walkthrough

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#### Step 1 — Create a Resource Group with Tags

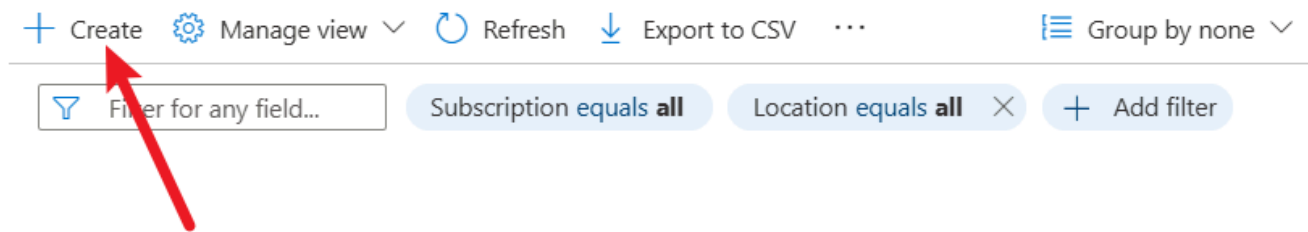
Signed into the Azure Portal and searched for **Resource Groups** to begin provisioning a tagged development environment for CloudFirst Retail.



Clicked **Create** to begin.

+ Create ⚙️ Manage view ▾ ↻ Refresh ⬇️ Export to CSV ... 📑 Group by none ▾

🔍 Filter for any field... Subscription equals all Location equals all ✕ + Add filter



## No resource groups to display

Resource groups provide a logical container to manage and organize Azure resources, simplifying administration and enabling efficient resource management.

+ Create



[Learn more](#)

Showing 1 - 0 of 0. Display count: auto ▾

 [Give feedback](#)

Selected the subscription, entered the Resource Group name `rg-cloudfirst-dev`, and chose **East US** as the region.

portal.azure.com/#view/HubsExtension/ResourceGroupCr...

Microsoft Azure Search resources, services, and docs (G+/) Copilot

Home > Resource Manager | Resource groups

## Create a resource group

Basics Tags Review + create

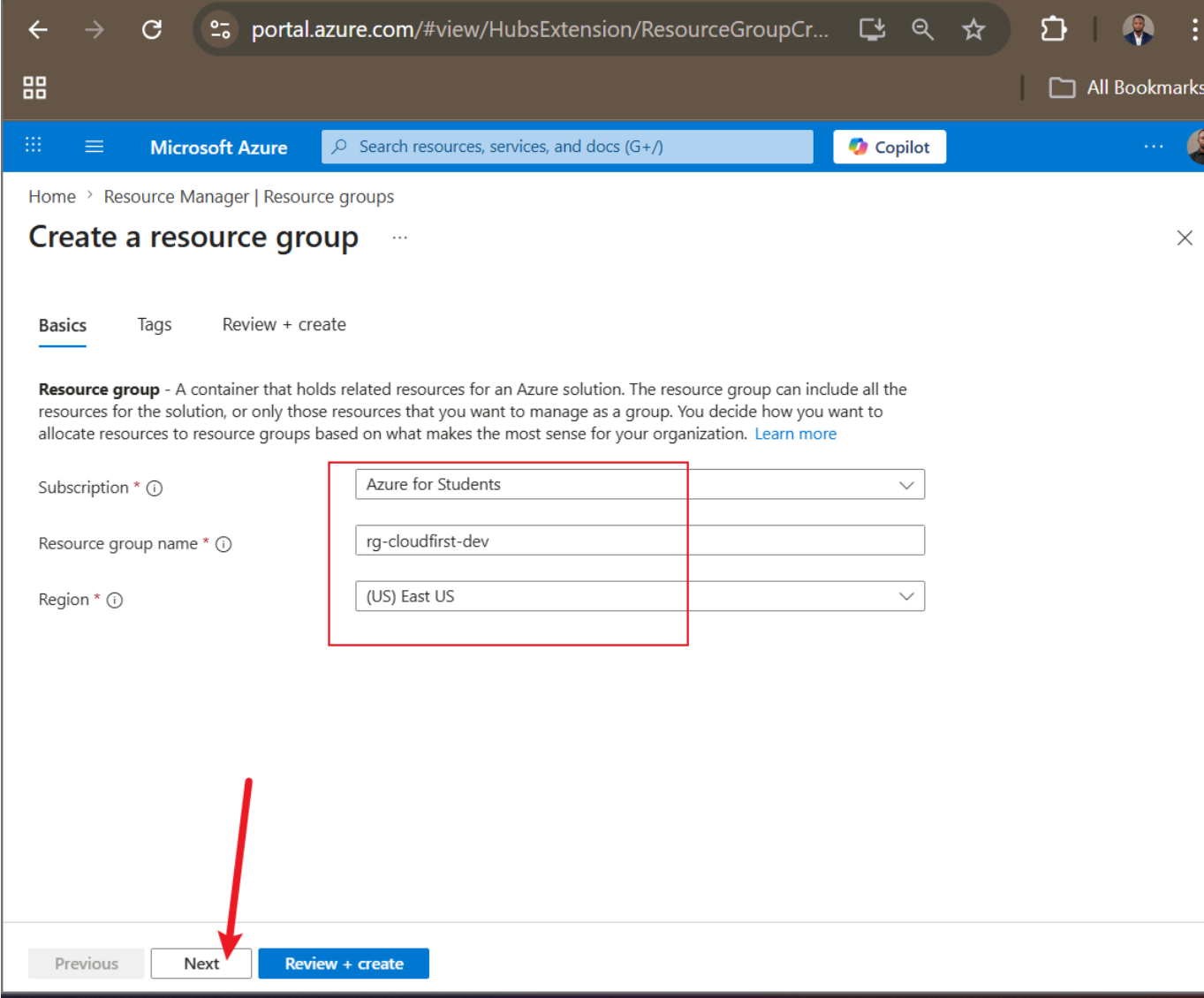
**Resource group** - A container that holds related resources for an Azure solution. The resource group can include all the resources for the solution, or only those resources that you want to manage as a group. You decide how you want to allocate resources to resource groups based on what makes the most sense for your organization. [Learn more](#)

Subscription \* ⓘ Azure for Students

Resource group name \* ⓘ rg-cloudfirst-dev

Region \* ⓘ (US) East US

Previous Next Review + create



Added the following governance tags to enable cost tracking and resource management:

| Tag         | Value             |
|-------------|-------------------|
| Environment | Development       |
| Project     | CloudFirst-Retail |
| CostCenter  | IT-Development    |

The screenshot shows the Microsoft Azure portal interface. The browser address bar displays `portal.azure.com/#view/HubsExtension/ResourceGroupCr...`. The page title is "Create a resource group". The navigation tabs are "Basics", "Tags" (which is selected), and "Review + create".

Below the tabs, there is a descriptive text: "Apply tags to your Azure resources to logically organize them by categories. A tag consists of a key (name) and a value. Tag names are case-insensitive and tag values are case-sensitive. [Learn more](#)".

A table is displayed with the following columns: "Name", "Value", and "Resource".

| Name        | Value | Resource          |                |  |
|-------------|-------|-------------------|----------------|--|
| Environment | :     | Development       | Resource group |  |
| Project     | :     | CloudFirst-Retail | Resource group |  |
| CostCenter  | :     | IT-Development    | Resource group |  |
|             | :     |                   | Resource group |  |

At the bottom of the page, there are three buttons: "Previous", "Next", and "Review + create". A red arrow points to the "Review + create" button.

portal.azure.com/#view/HubsExtension/ResourceGroupCr...

Microsoft Azure Search resources, services, and docs (G+/) Copilot

Home > Resource Manager | Resource groups

## Create a resource group ...

Basics Tags Review + create

[Automation Link](#)

### Basics

|                     |                    |
|---------------------|--------------------|
| Subscription        | Azure for Students |
| Resource group name | rg-cloudfirst-dev  |
| Region              | East US            |

### Tags

|             |                   |
|-------------|-------------------|
| Environment | Development       |
| Project     | CloudFirst-Retail |
| CostCenter  | IT-Development    |

Previous Next **Create**

Resource Group created successfully. Clicked to open the RG dashboard.

portal.azure.com/#servicemenu/Microsoft\_Azure\_Resource\_...

Microsoft Azure Search resources, services, and docs (G+/) Copilot

Home > Resource Manager

## Resource Manager | Resource groups

Default Directory

Search

Resource Manager

All resources

Favorite resources

Recent resources

**Resource groups**

Tags

> Organization

> Tools

> Deployments

Templates

Template specs

> Help

+ Create Manage view Refresh Export

Filter for any field...

Subscription equals all Location equals all + Add filter

**Resource group created**

Creating resource group 'rg-cloudfirst-dev' in subscription 'Azure for Students' succeeded.

Go to resource gr... Pin to dashbo...

### No resource groups to display

Resource groups provide a logical container to manage and organize Azure resources, simplifying administration and enabling efficient resource management.

+ Create

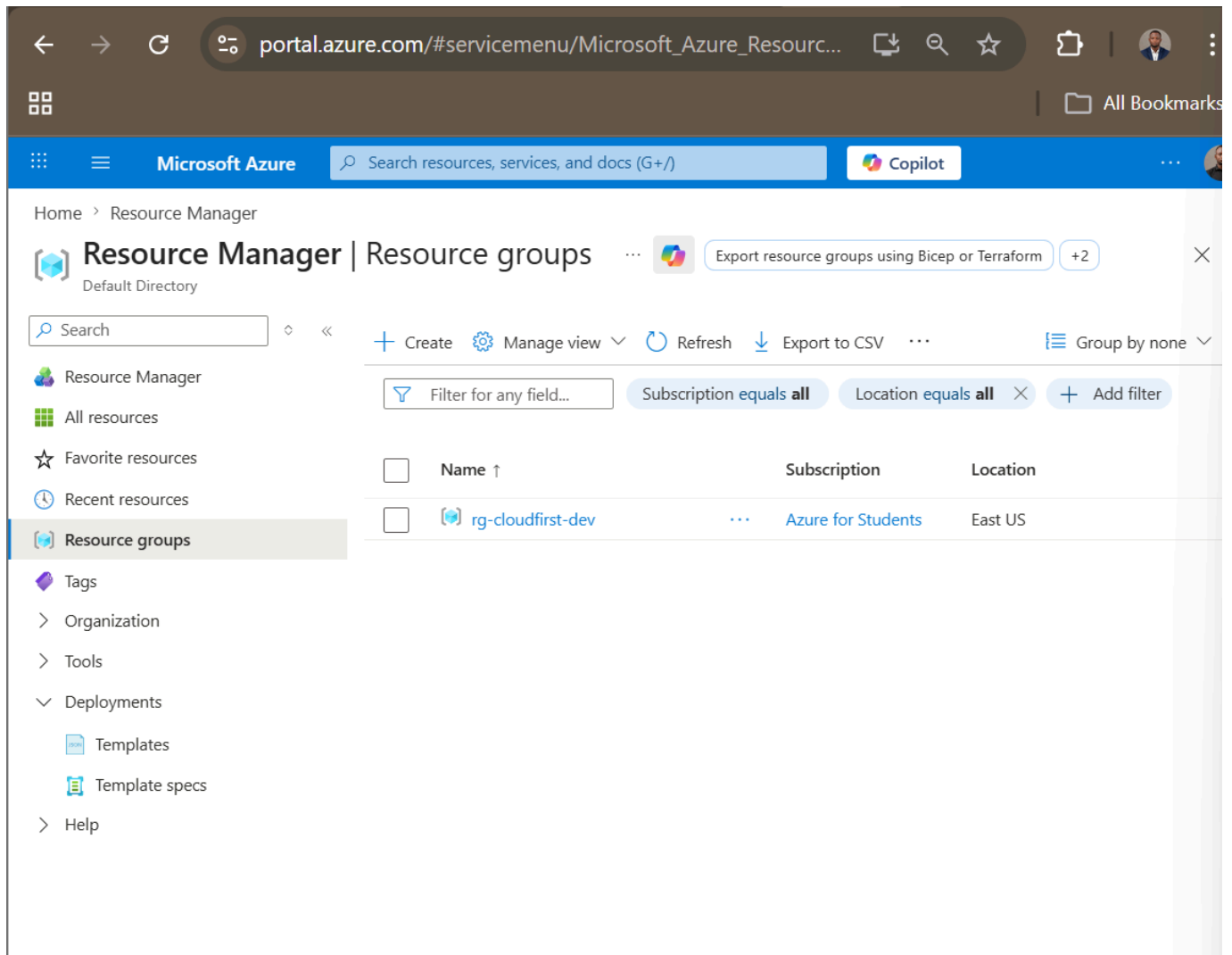
[Learn more](#)

Showing 1 - 0 of 0. Display count: auto

[Give feedback](#)

Add or remove favorites by pressing Ctrl+Shift+F





Home > Resource Manager

## Resource Manager | Resource groups

Default Directory

Search resources, services, and docs (G+/)

Export resource groups using Bicep or Terraform +2

Search

+ Create Manage view Refresh Export to CSV Group by none

Filter for any field...

Subscription equals all Location equals all Add filter

| <input type="checkbox"/> | Name ↑            |     | Subscription       | Location |
|--------------------------|-------------------|-----|--------------------|----------|
| <input type="checkbox"/> | rg-cloudfirst-dev | ... | Azure for Students | East US  |

Resource Manager

All resources

Favorite resources

Recent resources

Resource groups

Tags

Organization

Tools

Deployments

Templates

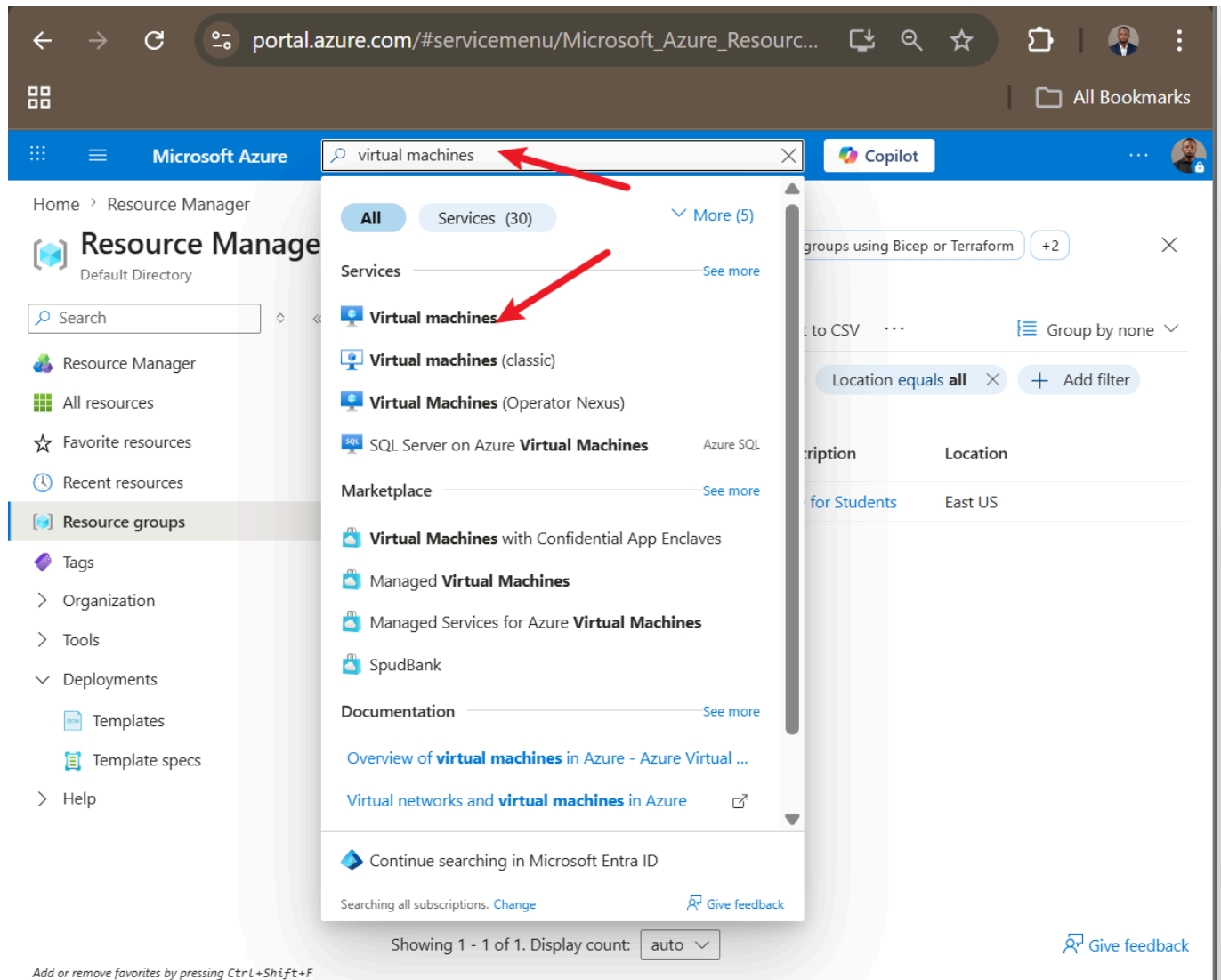
Template specs

Help

**Key Insight:** Tags are metadata applied to resources. They enable cost tracking by department, automated governance via Azure Policy, and compliance reporting. Azure resources support up to **50 tags** each.

## Step 2 — Explore Virtual Machine Configuration (Without Deploying)

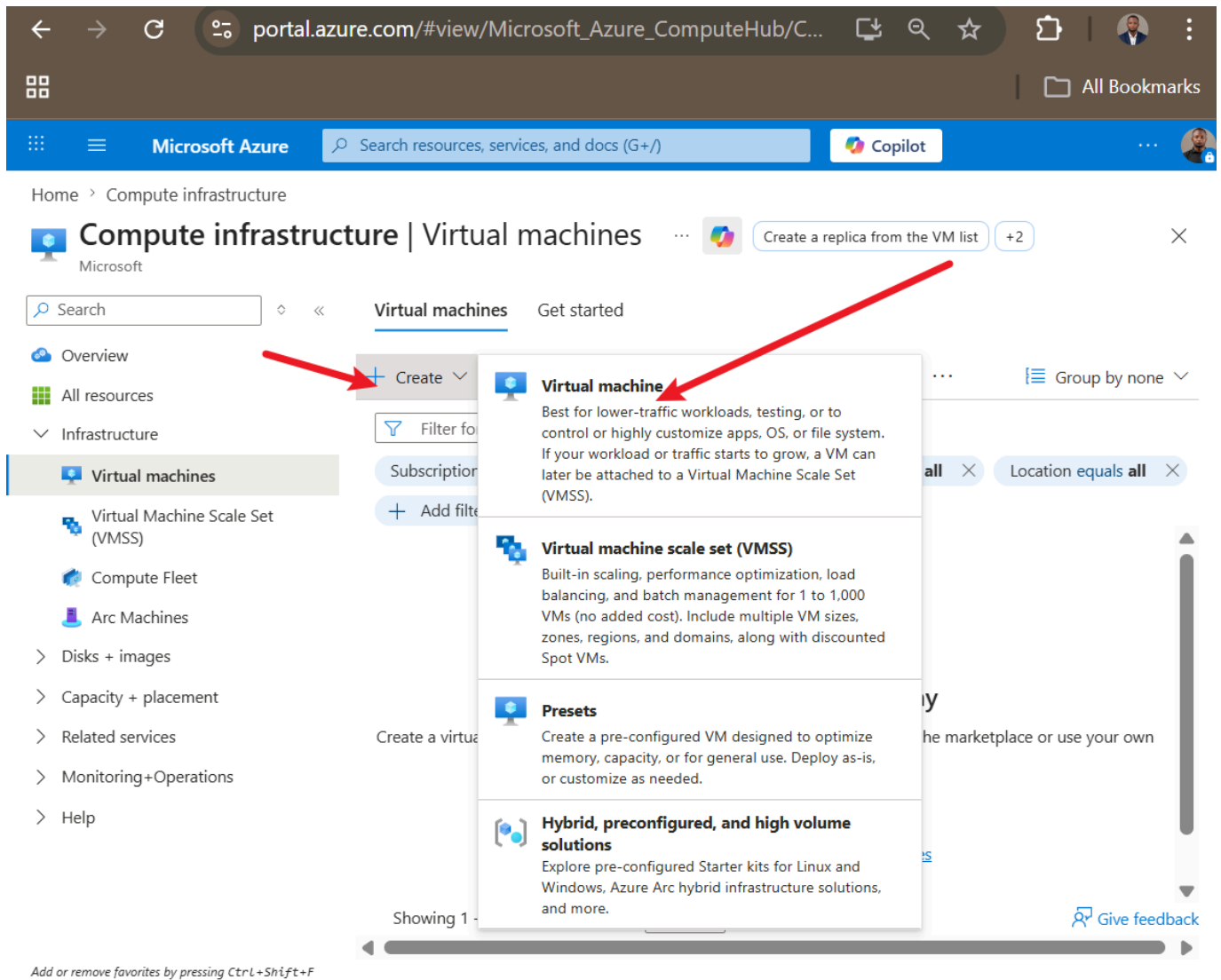
Searched for **Virtual Machines** in the portal search bar and selected it.



Clicked **Create**.

The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Azure logo, a search bar, and a Copilot button. The main header indicates the current location: Home > Compute infrastructure. The page title is 'Compute infrastructure | Virtual machines'. A search bar is present, and a button 'Create a replica from the VM list' is visible. The left sidebar shows the navigation menu with 'Virtual machines' selected. The main content area displays a message 'No virtual machines to display' and a 'Create' button, which is highlighted with a red arrow. Below the message, there are links for 'Learn more about Windows virtual machines' and 'Learn more about Linux virtual machines'. The bottom of the page shows a status bar with 'Showing 1 - 0 of 0. Display count: auto' and a 'Give feedback' button.

Selected **Virtual machine** from the dropdown.



## Basics Tab — Project Details

Selected the subscription and `rg-cloudfirst-dev` as the Resource Group.

Create a virtual machine ...

Basics | Disks | Networking | Management | Monitoring | Advanced | Tags | Review + create

Create a virtual machine that runs Linux or Windows. Select an image from Azure marketplace or use your own customized image. Complete the Basics tab then Review + create to provision a virtual machine with default parameters or review each tab for full customization. [Learn more](#)

This subscription may not be eligible to deploy VMs of certain sizes in certain regions.

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription \*

Azure for Students

This virtual machine requires a subscription without any spending limit or temporary payment method set. Use a different subscription or update your subscription (requires account administrator privileges) at [\[Account Center\]](#) (<https://account.windowsazure.com>) to continue.

Resource group \*

rg-cloudfirst-dev

Create new

Instance details

< Previous

Next : Disks >

Review + create

Give feedback

Configured instance details:

- **VM name:** vm-cloudfirst-dev
- **Region:** East US
- **Availability options:** No infrastructure redundancy required
- **Security type:** Standard
- **Image:** Ubuntu Server 24.04 LTS - x64 Gen2

Create new

Instance details

Virtual machine name \*

vm-cloudfirst-dev

Region \*

(US) East US

[Deploy to an Azure Extended Zone](#)

Availability options

No infrastructure redundancy required

Security type

Standard

Image \*

Ubuntu Server 20.04 LTS - x64 Gen2

[See all images](#) | [Configure VM generation](#)

VM architecture

☐ Arm64

☒ x64

Run with Azure Spot discount

☐

Selected the **B1s** size (1 vCPU, 1 GiB RAM) at **\$7.59/month** — a Burstable, cost-effective instance ideal for dev/test variable workloads. Set authentication type to Password.

portal.azure.com/#view/Microsoft\_Azure\_Compute/C...

Microsoft Azure Search resources, services, and docs (G+/)

Copilot

Home > Compute infrastructure | Virtual machines

## Create a virtual machine

⚠ Changing Basic options may reset selections you have made. Review all options prior to creating the virtual machine.

Run with Azure Spot discount ☐

Size \*

[See all sizes](#)

✖ This size is currently unavailable in eastus for this subscription: NotAvailableForSubscription.

[Request quota](#)

[Refresh quota](#)

Enable Hibernation ☐

**Administrator account**

Authentication type ☐ SSH public key ☒ Password

[Learn more](#)

< Previous Next : Disks > Review + create

[Give feedback](#)

Set the admin username, password, and configured inbound ports to allow **SSH (port 22)**.

portal.azure.com/#view/Microsoft\_Azure\_Compute/C...

Microsoft Azure Search resources, services, and docs (G+/) Copilot

Home > Compute infrastructure | Virtual machines

## Create a virtual machine ...

⚠ Changing Basic options may reset selections you have made. Review all options prior to creating the virtual machine.

Username \* ⓘ azureuser ✓

Password \* ..... ✓

Confirm password \* ..... ✓

### Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

Public inbound ports \* ⓘ ☐ None ☒ Allow selected ports

Select inbound ports \* SSH (22) ✓

⚠ This will allow all IP addresses to access your virtual machine. This is only recommended for testing. Use the Advanced controls in the Networking tab to create rules to limit inbound traffic to known IP addresses.

< Previous Next : Disks > Review + create Give feedback

⚠ **Security Note:** Opening SSH port 22 to the internet is flagged with a security warning. In production, use Azure Bastion or restrict source IPs via Network Security Group rules.

## Disks Tab

Selected **Standard SSD (locally redundant storage)** — a balanced cost/performance option at ~\$6–8/month, recommended for most workloads. Proceeded to Networking.

portal.azure.com/#view/Microsoft\_Azure\_Compute/C...

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

Home > Compute infrastructure | Virtual machines

Create a virtual machine

Azure disk storage encryption automatically encrypts your data stored on Azure managed disks (OS and data disks) at rest by default when persisting it to the cloud.

Encryption at host 

Encryption at host is not registered for the selected subscription. [Learn more](#)

OS disk

OS disk size 

Image default

OS disk type \* 

Standard SSD (locally-redundant storage)

The selected VM size supports premium disks. We recommend Premium SSD for high IOPS workloads. Virtual machines with Premium SSD disks qualify for the 99.9% connectivity SLA.

Delete with VM 

☒

Key management 

Platform-managed key

Enable Ultra Disk compatibility 

☐

Ultra disk is supported in Availability Zone(s) 1,2,3 for the selected VM size Standard\_B1s.

< Previous

Next : Networking >

Review + create

Give feedback

Disk Type vs. SLA:

| Disk Type    | ~Cost/Month | Single-Instance SLA |
|--------------|-------------|---------------------|
| Standard HDD | \$4–5       | No SLA              |
| Standard SSD | \$6–8       | No SLA              |
| Premium SSD  | \$20–25     | 99.9%               |

Networking Tab

Observed that Azure **automatically generates** all networking resources — Virtual Network, subnet, Public IP, and Network Security Group — when a VM is configured. No manual setup required.



portal.azure.com/#view/Microsoft\_Azure\_Compute/C... | All Bookmarks

Microsoft Azure Search resources, services, and docs (G+/) Copilot

Home > Compute infrastructure | Virtual machines

## Create a virtual machine

Help me create a VM optimized for high availability +2

Virtual network \* ⓘ (new) vm-cloudfirst-dev -vnet  
[Create new](#)

Subnet \* ⓘ (new) default (10.0.0.0/24)

Public IP ⓘ (new) vm-cloudfirst-dev -ip  
[Create new](#)  
**i** Public IP addresses have a nominal charge. [Estimate price](#)

NIC network security group ⓘ  
☐ None  
☒ Basic  
☐ Advanced

Public inbound ports \* ⓘ  
☐ None  
☒ Allow selected ports

Select inbound ports \* SSH (22)

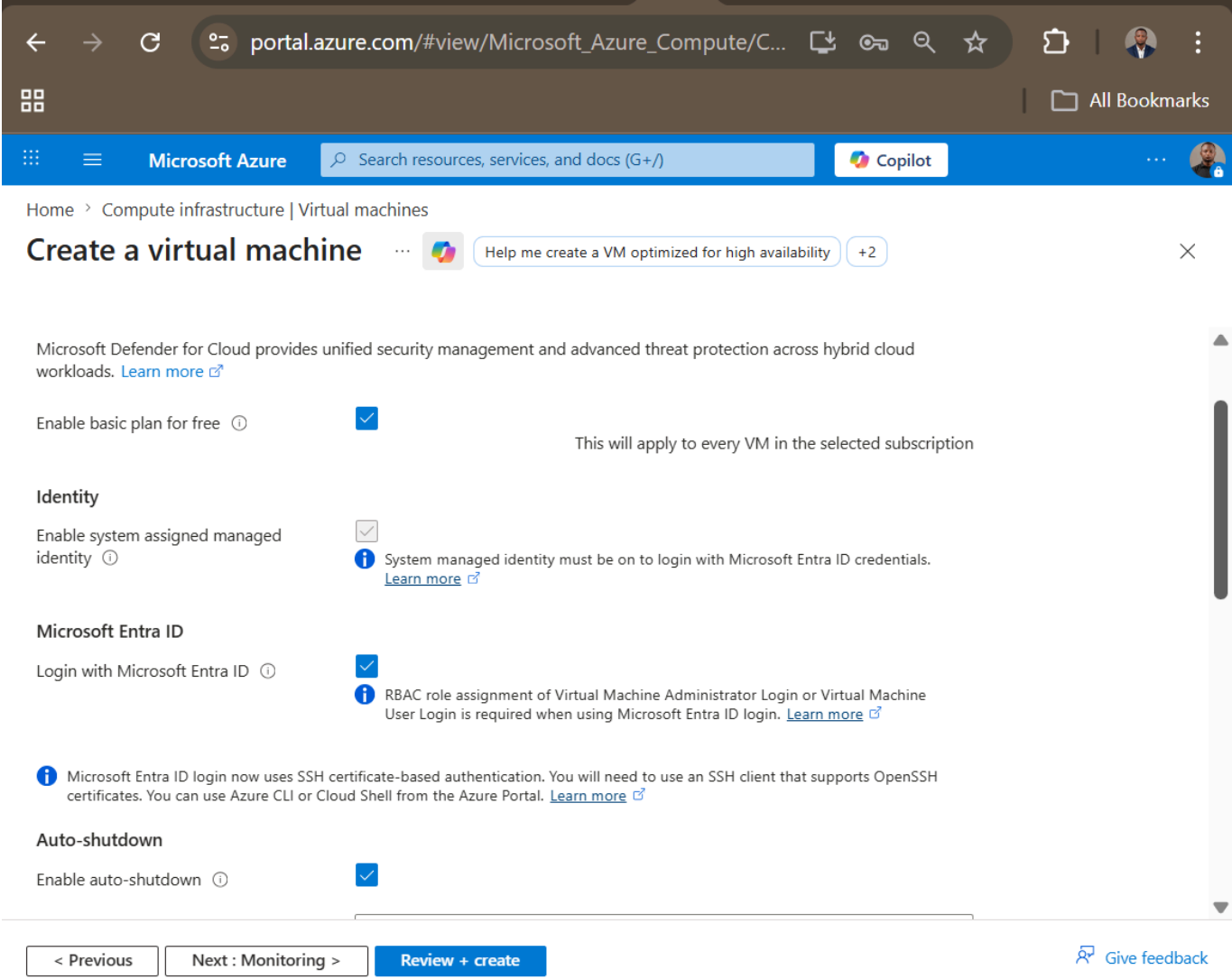
**⚠ This will allow all IP addresses to access your virtual machine.** This is only recommended for testing. Use the Advanced controls in the Networking tab to create rules to limit inbound traffic to known IP addresses.

< Previous Next : Management > Review + create Give feedback

**Key Insight:** VM deployment creates 6 resources automatically — the VM, VNet, subnet, NIC, Public IP, NSG, and OS disk. Each is a separately managed resource.

## Management Tab

Enabled **Managed Identity** and **Microsoft Entra ID** login, and noted the **Auto-shutdown** option — a cost-saving feature that schedules automatic VM shutdown at a specified time.



Enabled **Backup** and **Guest OS periodic assessment** for patch management.

portal.azure.com/#view/Microsoft\_Azure\_Compute/C...

Microsoft Azure Search resources, services, and docs (G+/) Copilot

Home > Compute infrastructure | Virtual machines

## Create a virtual machine

Help me create a VM optimized for high availability +2

Enable backup ☒

Recovery Services vault \* (new) defaultVault632 [Create new](#)

Policy subtype \* ☐ Standard ☒ Enhanced

- ✓ Once a day backup
- ✓ 1-5 days operational tier
- ✓ Vault tier
- ZRS resilient snapshot tier
- Support for Trusted launch and Confidential Azure VMs

- ✓ Multiple backups per day
- ✓ 1-30 days operational tier
- ✓ Vault tier
- ✓ ZRS resilient snapshot tier
- ✓ Support for Trusted launch and Confidential Azure VMs

Backup policy \* (new) EnhancedPolicy-mola4da8 [Create new](#)

Guest OS updates

Enable periodic assessment ☒

Patch orchestration options Image default [Some patch orchestration options are not available for this image. Learn more](#)

< Previous Next: Monitoring > Review + create Give feedback

## Monitoring Tab

Reviewed monitoring options — enabled **OS guest diagnostics** and **Boot diagnostics** (uses managed storage at no extra cost).

The screenshot shows the 'Create a virtual machine' wizard in the Microsoft Azure portal, specifically the 'Monitoring' tab. The breadcrumb navigation at the top indicates the path: Home > Compute infrastructure | Virtual machines. The page title is 'Create a virtual machine', and there is a help button that says 'Help me create a VM optimized for high availability' with a '+2' icon. The tabs at the top are Basics, Disks, Networking, Management, Monitoring (selected), Advanced, Tags, and Review + create. The instructions state: 'Configure monitoring options for your VM.' The 'Alerts' section has a checkbox for 'Enable recommended alert rules' which is unchecked. The 'Diagnostics' section has a sub-section 'Boot diagnostics' with three radio button options: 'Enable with managed storage account (recommended)' (selected), 'Enable with custom storage account', and 'Disable'. Below these is an information icon and the text 'There is a nominal charge for the managed storage account.' The 'Enable OS guest diagnostics' checkbox is checked. The 'Diagnostics storage account' is set to '(new) rgcloudfirstdevdiag' with a dropdown arrow and a 'Create new' link below it. The 'Health' section has a checkbox for 'Enable application health monitoring' which is unchecked. At the bottom, there are three buttons: '< Previous', 'Next > Advanced >' (highlighted with a red arrow), and 'Review + create'. A 'Give feedback' link is in the bottom right corner.

portal.azure.com/#view/Microsoft\_Azure\_Compute/C...

Microsoft Azure Search resources, services, and docs (G+)

Copilot

Home > Compute infrastructure | Virtual machines

## Create a virtual machine

Help me create a VM optimized for high availability +2

Basics Disks Networking Management **Monitoring** Advanced Tags Review + create

Configure monitoring options for your VM.

**Alerts**

Enable recommended alert rules ☐

**Diagnostics**

Boot diagnostics ☒ Enable with managed storage account (recommended)  
☐ Enable with custom storage account  
☐ Disable  
There is a nominal charge for the managed storage account.

Enable OS guest diagnostics ☒

Diagnostics storage account \* (new) rgcloudfirstdevdiag  
[Create new](#)

**Health**

Enable application health monitoring ☐

< Previous Next > Advanced > Review + create

[Give feedback](#)

### Advanced Tab

Left all Advanced configuration options at their defaults — Extensions, Cloud-init scripts, and User data are optional features not required for this lab.

The screenshot shows the 'Create a virtual machine' wizard in the Microsoft Azure portal, specifically the 'Advanced' tab. The breadcrumb navigation at the top reads 'Home > Compute infrastructure | Virtual machines'. The main heading is 'Create a virtual machine', followed by a search bar with the text 'Help me create a VM optimized for high availability' and a '+2' button. Below the heading are tabs for 'Basics', 'Disks', 'Networking', 'Management', 'Monitoring', 'Advanced' (which is selected), 'Tags', and 'Review + create'. A sub-header states: 'Add additional configuration, agents, scripts or applications via virtual machine extensions or cloud-init.' The 'Extensions' section includes a link 'Select an extension to install'. The 'VM applications' section includes a link 'Select a VM application to install'. The 'Custom data and cloud init' section includes a text box for 'Custom data' and a link 'Learn more about custom data for VMs'. A red arrow points to the 'Next : Tags >' button at the bottom of the wizard. Other buttons include '< Previous', 'Review + create', and 'Give feedback'.

## Tags Tab

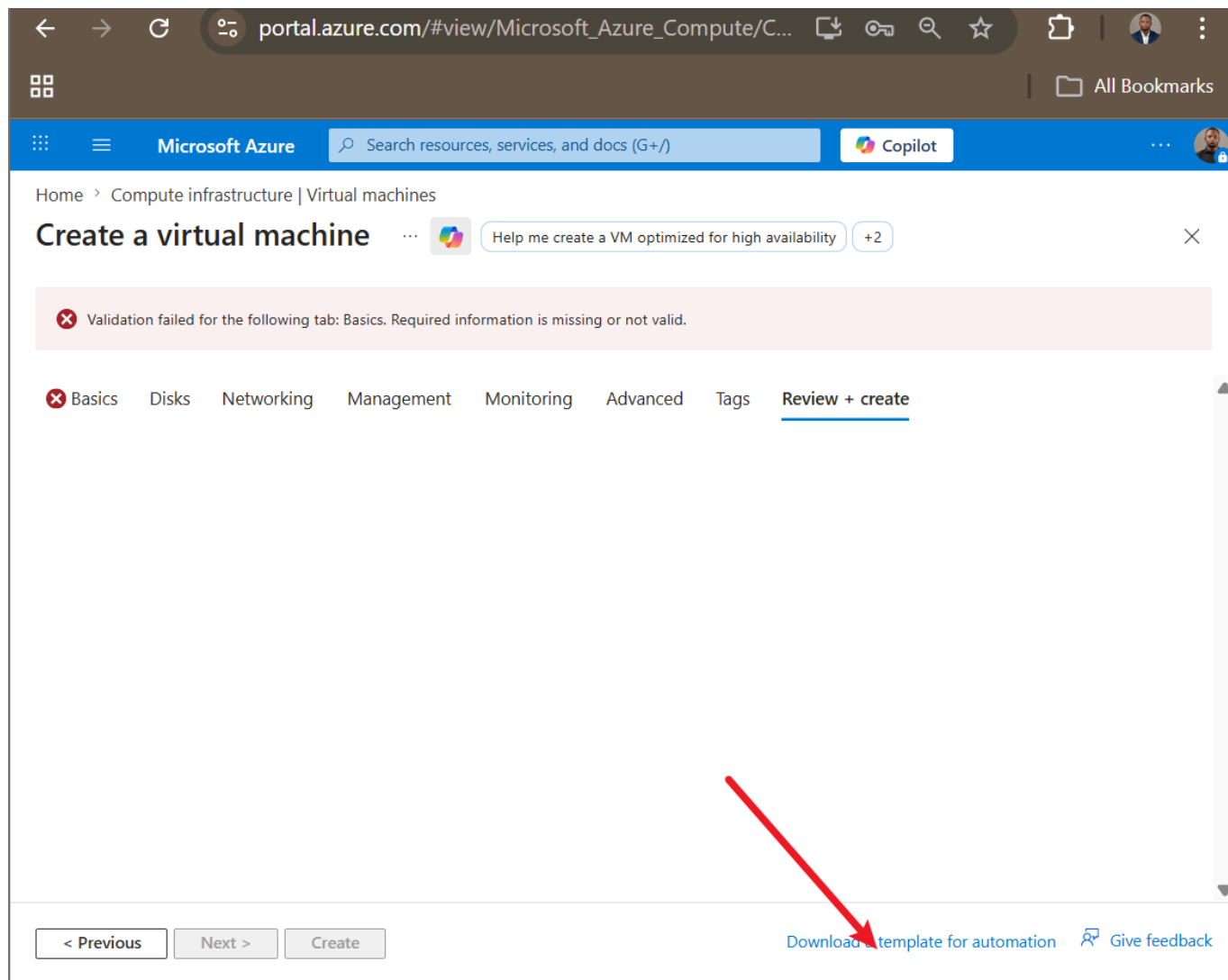
Manually applied the same three tags from the Resource Group to the VM. This is a critical exam concept — tags applied to the RG are **not inherited** by child resources.

 Add tags manually — not inherited from RG

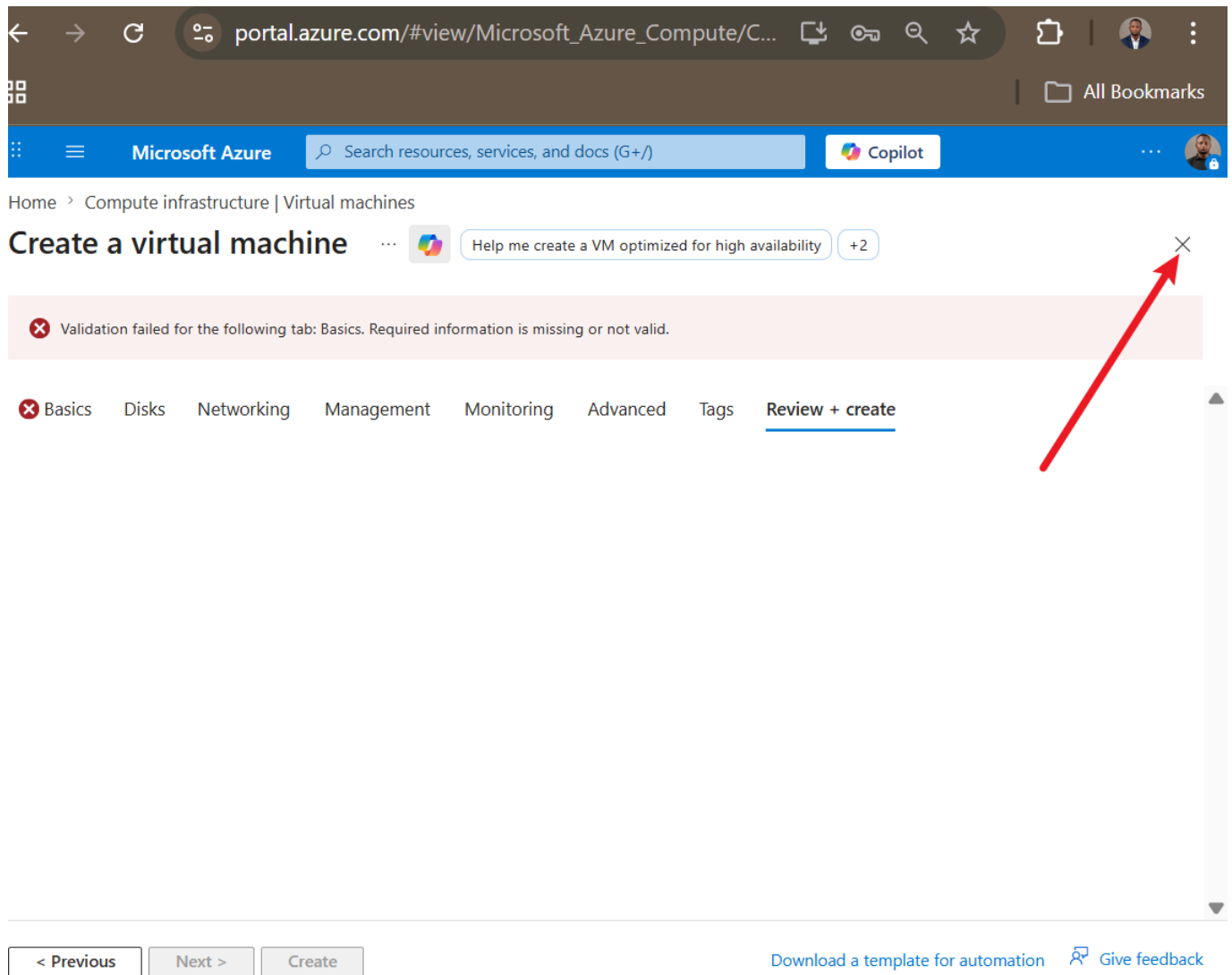
**⚠ Critical Exam Concept:** Tags do NOT inherit from Resource Groups to resources. RBAC permissions, Azure Policy assignments, and Resource Locks DO inherit — but tags must be applied explicitly at each resource level.

## Review + Create — Download ARM Template (No Deployment)

Ignored the validation error (expected for a conceptual lab) and clicked **Download a template for automation** to capture the ARM template — a JSON representation of the full VM configuration ready for IaC reuse.



Clicked **Cancel** — the full configuration was explored without deploying any resources or incurring costs.



### Step 3 — Compare VM Sizes & Pricing with Azure Pricing Calculator

Navigated to the **Azure Pricing Calculator** and added Virtual Machines to begin a three-way cost comparison across VM sizing families.

azure.microsoft.com/en-us/pricing/calculator/

Your Esti... [Log in](#) \$0.00 / Estimated upfront cost \$0.00 / Estimated monthly cost

[Log in](#) to save cost estimates and use your Azure agreement pricing.

Products Estimate templates Saved estimates FAQs

Select a product to include it in your estimate.

Virtual Machine

**Virtual Machines** ⓘ  
Provision Windows and Linux VMs in seconds  
[Add to estimate](#)

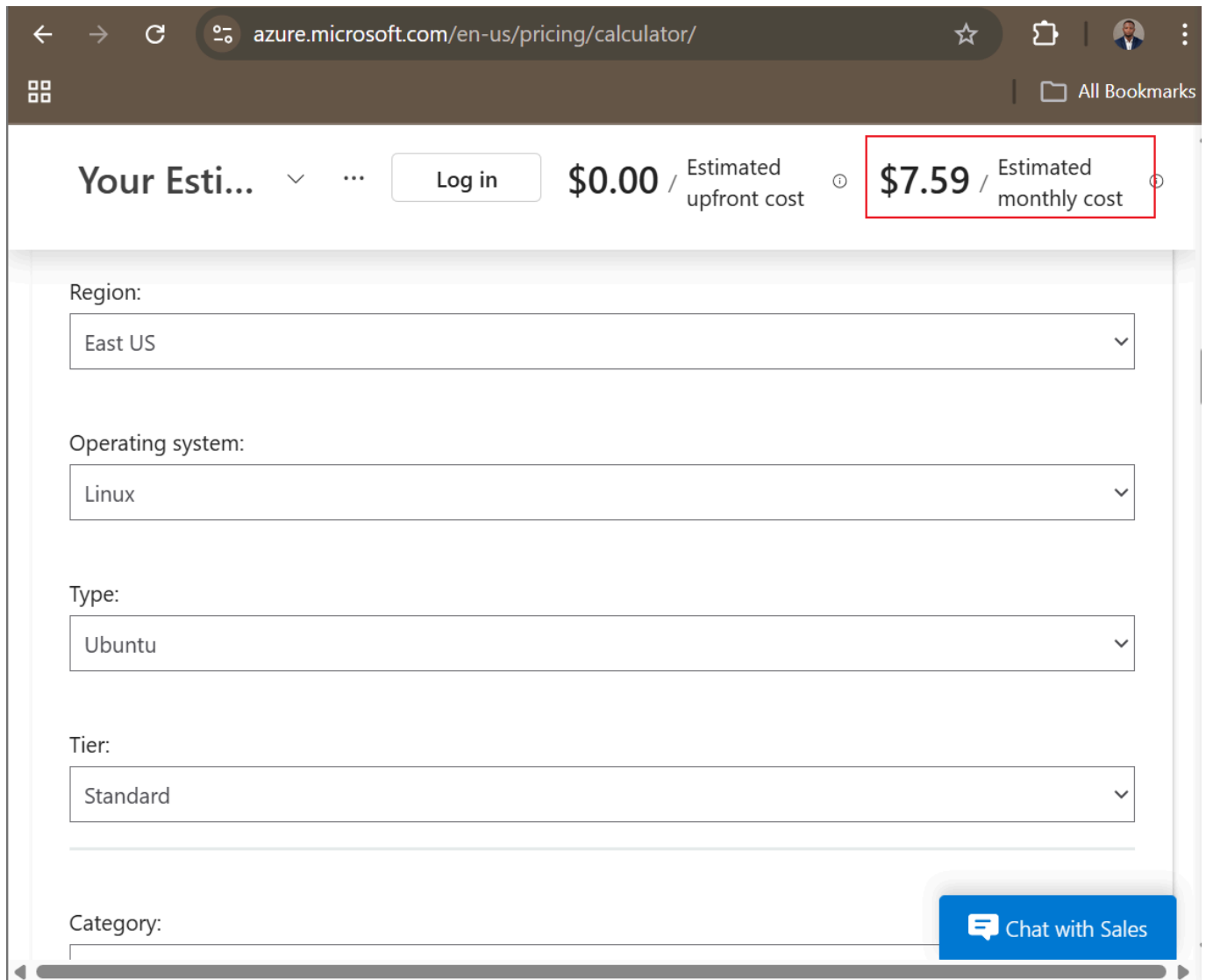
**Virtual Machine Scale Sets** ⓘ  
Manage and scale up to thousands of Linux and Windows VMs

[Chat with Sales](#)

### VM 1 — Development (B1s): \$7.59/month

Configured: Linux · Ubuntu · East US · Standard · **B1s (1 vCPU, 1 GiB RAM)**





The screenshot shows the Azure Pricing Calculator interface. The browser address bar displays `azure.microsoft.com/en-us/pricing/calculator/`. The top navigation bar includes a "Log in" button and two cost estimates: "\$0.00 / Estimated upfront cost" and "\$7.59 / Estimated monthly cost", with the latter highlighted by a red box. The main configuration area contains several dropdown menus: "Region:" set to "East US", "Operating system:" set to "Linux", "Type:" set to "Ubuntu", and "Tier:" set to "Standard". A "Category:" dropdown is partially visible at the bottom. A blue "Chat with Sales" button is located in the bottom right corner of the configuration area.

## VM 2 — Production (D2s v3): \$70.08/month

Added a second VM instance to the estimate.

The screenshot shows the Azure Pricing Calculator interface. At the top, the browser address bar displays `azure.microsoft.com/en-us/pricing/calculator/`. The page header includes a 'Log in' button and cost estimates: '\$0.00 / Estimated upfront cost' and '\$7.59 / Estimated monthly cost'. Below the header, a navigation bar contains 'Products', 'Estimate templates', 'Saved estimates', and 'FAQs'. A blue banner prompts the user to 'Select a product to include it in your estimate.' A search bar contains the text 'Virtual Machine'. Below the search bar, four product cards are displayed: 'Virtual Machines' (with a red arrow pointing to its 'Add to estimate' button), 'Virtual Machine Scale Sets', 'Azure Dedicated Host', and 'IP Addresses'. Each card includes a brief description and an 'Add to estimate' button. A 'Chat with Sales' button is visible in the bottom right corner.

Configured: Linux · Ubuntu · East US · Standard · **D2s v3 (2 vCPU, 8 GiB RAM)**

azure.microsoft.com/en-us/pricing/calculator/

Your Esti... Log in **\$0.00** / Estimated upfront cost **\$77.67** / Estimated monthly cost

Tier:  
Standard

Category:  
All

Instance Series:  
All

INSTANCE: [\(Need help finding the right VM?\)](#)  
B1s: 1 Cores, 1 GB RAM, 4 GB Temporary storage, \$0.010/hour

[Chat with Sales](#)

**Production VM estimated monthly cost: \$70.08** — approximately 9× the cost of the dev B1s.

Your Esti...

Log in

\$0.00 / Estimated upfront cost

\$77.67 / Estimated monthly cost

3 year reserved (~62% discount)

\$70.08

Average per month (\$0.00 charged upfront)

= \$70.08

Average per month (\$0.00 charged upfront)

Managed Disks

\$0.00

Storage transactions

\$0.00

Bandwidth

\$0.00

Upfront cost

\$0.00

Monthly cost

\$70.08

Chat with Sales

VM 3 — Memory-Optimized (E2s v3): \$91.98/month

Added a third comparison instance: **E2s v3 (2 vCPU, 16 GiB RAM)** for memory-intensive workloads such as databases and caching.

←→↻🔍 azure.microsoft.com/en-us/pricing/calculator/🌟📁👤⋮

☐☐☐☐ | 📁 All Bookmarks

Your Esti... ▾⋮

Log in

\$0.00 / Estimated upfront cost ⓘ

\$169.65 / Estimated monthly cost ⓘ

Instance Series:

All ▾

INSTANCE: [\(Need help finding the right VM?\)](#)

🔍 E2s v3: 2 vCPUs, 16 GB RAM, 32 GB Temporary storage, \$0.126/hour ▾

1 × 730 Hours ▾

Virtual machines

Savings Options

Explore pricing models to help optimize your Azure costs. [Learn more](#)

Compute (E2s v3)

☒ Pay as you go

💬 Chat with Sales

Memory-optimized VM estimated monthly cost: \$91.98

Your Esti...

Log in

\$0.00

Estimated upfront cost

\$169.65

Estimated monthly cost

\$91.98

Average per month (\$0.00 charged upfront)

\$91.98

Average per month (\$0.00 charged upfront)

Managed Disks

\$0.00

Storage transactions

\$0.00

Bandwidth

\$0.00

Upfront cost

\$0.00

Monthly cost

\$91.98

Support

Chat with Sales

Exploring Savings Plans and Reserved Instances

Explored all available commitment-based pricing options to understand long-term savings:

1-Year Savings Plan — 31% discount → \$63.47/month

Your Esti...

▼

...

Log in

\$0.00

/ Estimated upfront cost ⓘ

\$141.14

/ Estimated monthly cost ⓘ

☒ 1 year savings plan  
(~31% discount)

☐ 3 year savings plan  
(~53% discount)

Compute payment options:

Monthly▼

Reservations ⓘ

☐ 1 year reserved  
(~38% discount)

☐ 3 year reserved  
(~60% discount)

\$63.47  
Average per month  
(\$0.00 charged upfront)

=

\$63.47

Chat with Sales

**3-Year Savings Plan — 53% discount → \$43.23/month**

All Bookmarks

Your Esti...

▼

...

Log in

\$0.00

/ Estimated upfront cost

ⓘ

\$120.90

/ Estimated monthly cost

ⓘ

☐ 1 year savings plan (~31% discount)

☒ 3 year savings plan (~53% discount)

Compute payment options:

Monthly

▼

Reservations ⓘ

☐ 1 year reserved (~38% discount)

☐ 3 year reserved (~60% discount)

\$43.23

Average per month (\$0.00 charged upfront)

= \$43.23

Chat with Sales

1-Year Reserved Instance — 38% discount → \$57.08/month

azure.microsoft.com/en-us/pricing/calculator/

☆

All Bookmarks

Your Estimate

▼

...

Log in

\$0.00

/ Estimated upfront cost

ⓘ

\$134.76

/ Estimated monthly cost

ⓘ

Savings plan ⓘ

☐ 1 year savings plan (~31% discount)

☐ 3 year savings plan (~53% discount)

Reservations ⓘ

☒ 1 year reserved (~38% discount)

☐ 3 year reserved (~60% discount)

Compute payment options:

Monthly

▼

\$57.08

Average per month (\$0.00 charged upfront)

= \$57.08

Average per month (\$0.00 charged upfront)

Managed Disks

Chat with Sales

3-Year Reserved Instance — 60% discount → \$36.05/month



☰

All Bookmarks

Your Estimate ▾ ...

Log in

\$0.00 / Estimated upfront cost ⓘ

\$114.17 / Estimated monthly cost ⓘ

Savings plan ⓘ

☐ 1 year savings plan (~31% discount)

☐ 3 year savings plan (~53% discount)

Reservations ⓘ

☐ 1 year reserved (~38% discount)

☒ 3 year reserved (~60% discount)

Compute payment options:

Monthly ▾

\$36.50

Average per month

(\$0.00 charged upfront)

=

\$36.50

Average per month

(\$0.00 charged upfront)

✓ Managed Disks

Chat with Sales

## VM Cost Comparison Summary

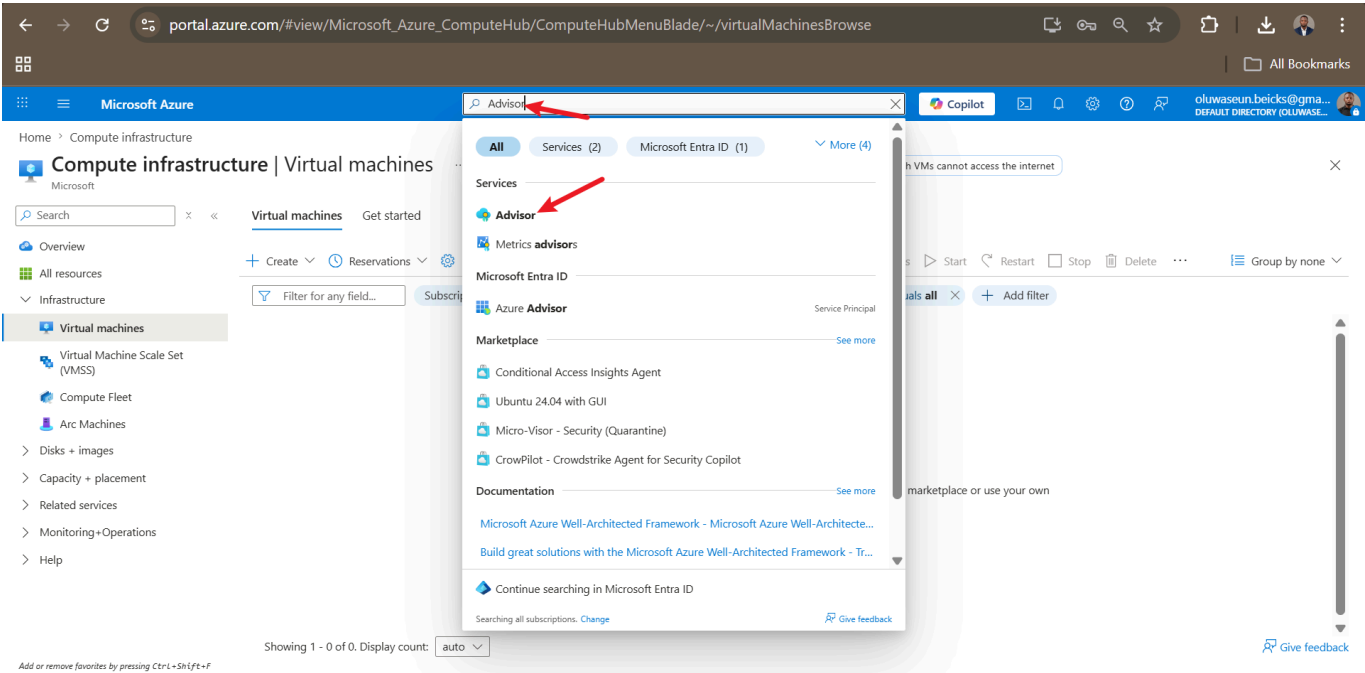
| VM     | Series                 | vCPU | RAM    | Pay-as-you-go | 1-Yr Reserved | 3-Yr Reserved |
|--------|------------------------|------|--------|---------------|---------------|---------------|
| B1s    | Burstable (Dev)        | 1    | 1 GiB  | \$7.59        | —             | —             |
| D2s v3 | General Purpose (Prod) | 2    | 8 GiB  | \$70.08       | \$57.08       | \$36.05       |
| E2s v3 | Memory-Optimized       | 2    | 16 GiB | \$91.98       | —             | —             |

### VM Series Quick Reference:

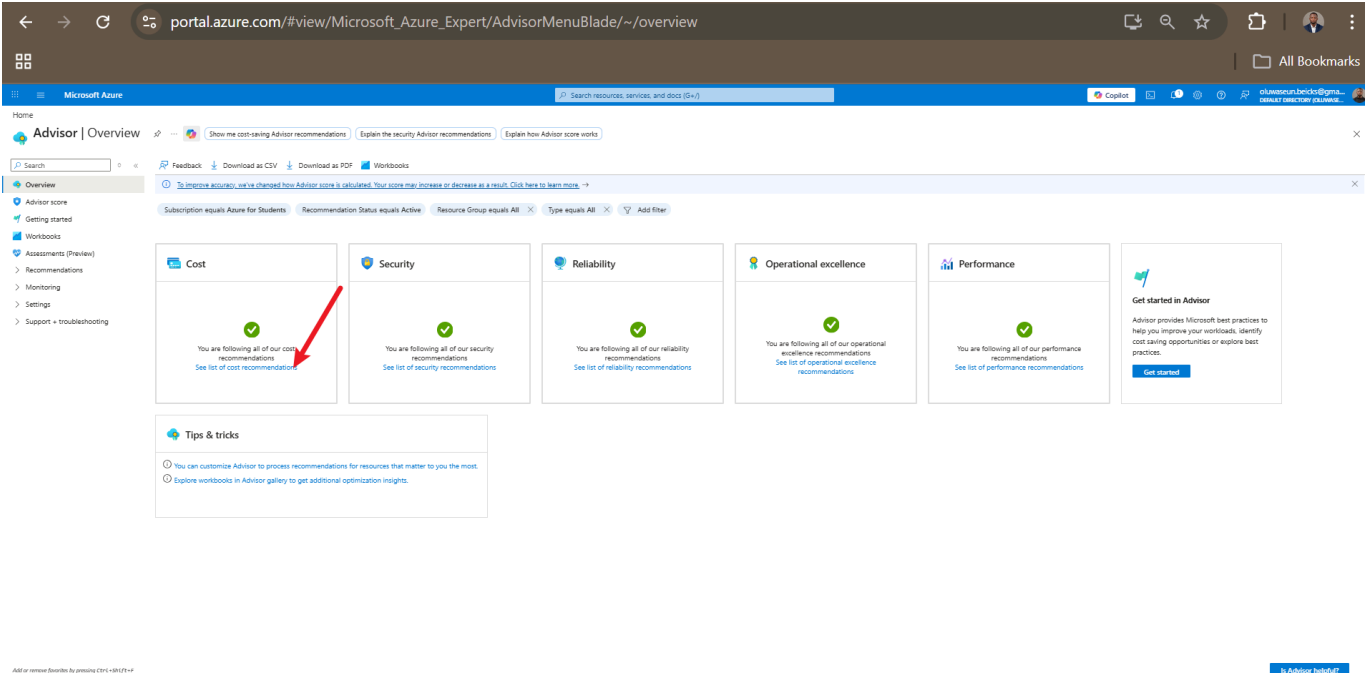
- **B = Burstable** — Budget dev/test, variable workloads
- **D = Daily Use** — Balanced CPU/memory, production apps
- **E = Enormous Memory** — RAM-intensive databases, caching
- **F = Fast CPU** — Compute-intensive batch processing
- **M = Massive Memory** — Terabyte-scale configurations

## Step 4 — Explore Azure Advisor Recommendations

Switched back to the Azure Portal and searched for **Advisor**.



Reviewed the five Advisor recommendation pillars and clicked through to **Cost** — confirmed that the environment is following current recommendations (no active cost alerts for the new environment).



Azure Advisor — Five Pillars:

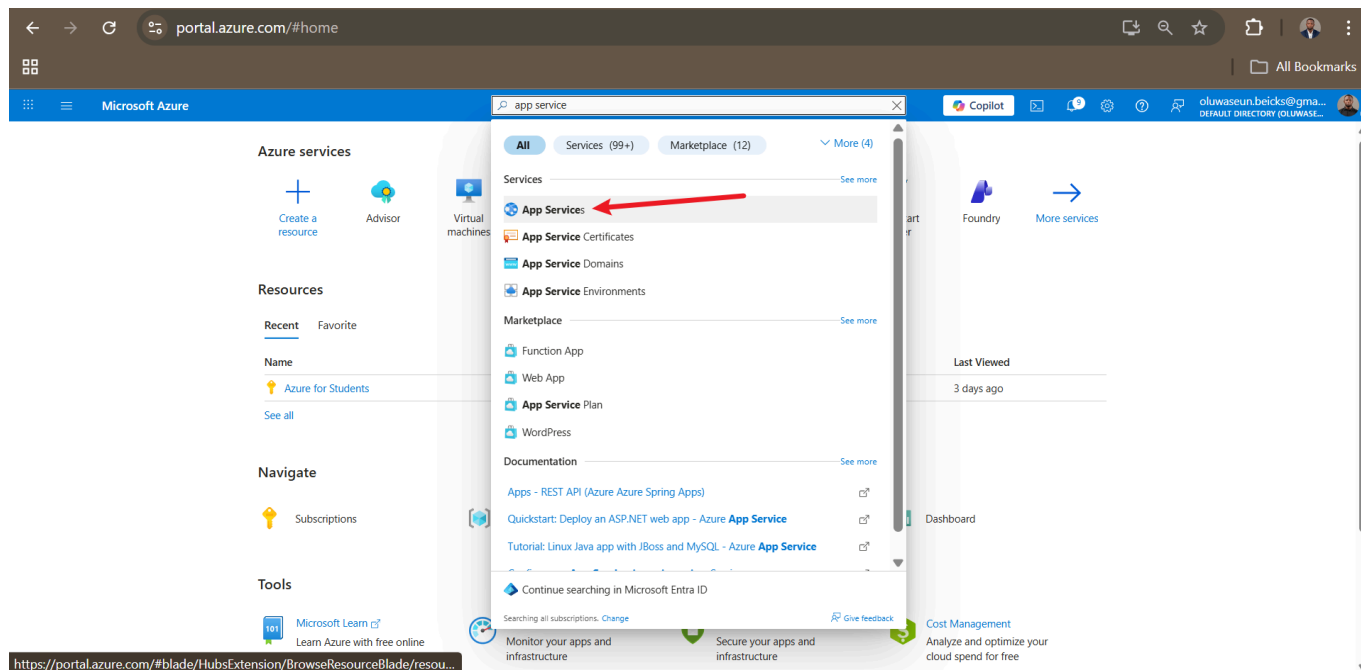
| Pillar                 | What It Addresses                                               |
|------------------------|-----------------------------------------------------------------|
| Cost                   | Right-size VMs, delete unused resources, buy Reserved Instances |
| Security               | Enable MFA, apply NSG rules, enable disk encryption             |
| Reliability            | Use Availability Zones, enable backup                           |
| Operational Excellence | Implement tagging, configure monitoring                         |
| Performance            | Upgrade disk types, scale resources                             |

Advisor is **completely free**, requires no configuration, and needs **24–48 hours** of usage data to generate active recommendations.

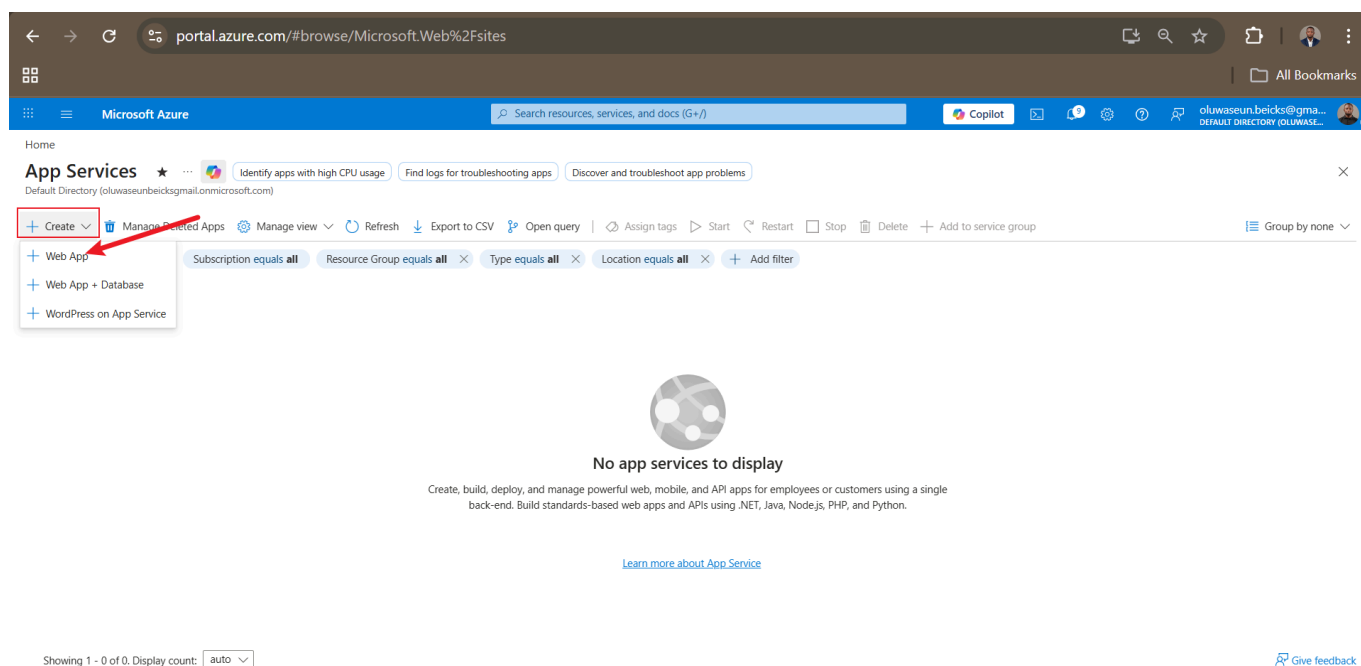
## Step 5 — Compare Azure Compute Service Options

### App Service (PaaS) — Explore Configuration

Searched for **App Services** and clicked on the result.



Clicked **Create** → **Web App**.



Configured basics: Subscription · **rg-cloudfirst-dev** · Name: **cf-dev-webapp** · Runtime: **Node 24** · OS: **Linux** · Region: **East US**.

The screenshot shows the 'Create Web App' blade in the Azure Portal. The 'Project Details' section is active, showing the following configuration:

- Subscription:** Azure for Students
- Resource Group:** rg-cloudfirst-dev (with a 'Create new' link)
- Instance Details:**
  - Name:** cf-dev-webapp (with a checkmark and '.azurewebsites.net' suffix)
  - Try a secure unique default hostname:** ☐ (disabled)
- Publish:** ☒ Code ☐ Container
- Runtime stack:** Node 24 LTS
- Operating System:** ☒ Linux ☐ Windows
- Region:** East US (with a note: 'Not finding your App Service Plan? Try a different region or select your App Service Environment.')

At the bottom, there are three buttons: 'Review + create' (highlighted in blue), '< Previous', and 'Next: Database >'.

**Key Observation vs. VM:** No OS selection screen, no disk configuration, no networking setup. Azure manages all infrastructure — you only configure the application runtime.

Clicked **Explore pricing plans** to review available tiers.

The screenshot shows the 'Pricing plans' section of the 'Create Web App' blade. The configuration is as follows:

- Linux Plan (East US):** (New) ASP-rgcloudfirstdev-a7f0 (with a 'Create new' link)
- Pricing plan:** Free F1 (Shared infrastructure) (with a red arrow pointing to the 'Explore pricing plans' link below it)

Below the pricing plan, there is a 'Zone redundancy' section with two radio buttons:

- ☐ **Enabled:** Your App Service plan and the apps in it will be zone redundant. The minimum App Service plan instance count will be two.
- ☒ **Disabled:** Your App Service plan and the apps in it will not be zone redundant. The minimum App Service plan instance count will be one.

At the bottom, there are three buttons: 'Review + create' (highlighted in blue), '< Previous', and 'Next: Database >'.

Clicked **Learn more about App Service pricing** to open the full pricing page.

portal.azure.com/#view/WebsitesExtension/ScaleSpecPicker.ReactView/data~/7B"allowAseV2Creation"%3Afalse%2C"allowAseV3Creation"...

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

oluwasun.beicks@gmail...  
DEFAULT DIRECTORY (OLUWASEUN...)

Home > App Services > Create Web App

Select App Service Pricing Plan

×

☒ Hardware view

☐ Feature view

Showing 31 App Service pricing plans

| Name             | ACU/vCPU               | vCPU | Memory (GB) | Remote Storage (GB) | Scale (instance) | SLA    | Cost per hour (instance) | Cost per month (instance) |
|------------------|------------------------|------|-------------|---------------------|------------------|--------|--------------------------|---------------------------|
| Popular options  |                        |      |             |                     |                  |        |                          |                           |
| Free F1          | 60 minutes/day compute | N/A  | 1           | 1                   | N/A              | N/A    |                          |                           |
| Basic B1         | 100                    | 1    | 1.75        | 10                  | 3                | 99.95% |                          |                           |
| Premium v4 P0V4  | -                      | 1    | 4           | 250                 | 30               | 99.95% |                          |                           |
| Premium v4 P1V4  | -                      | 2    | 8           | 250                 | 30               | 99.95% |                          |                           |
| Premium v4 P2V4  | -                      | 4    | 16          | 250                 | 30               | 99.95% |                          |                           |
| Premium v4 P3V4  | -                      | 8    | 32          | 250                 | 30               | 99.95% |                          |                           |
| Premium v4 P1mv4 | -                      | 2    | 16          | 250                 | 30               | 99.95% |                          |                           |
| Premium v4 P2mv4 | -                      | 4    | 32          | 250                 | 30               | 99.95% |                          |                           |
| Premium v4 P3mv4 | -                      | 8    | 64          | 250                 | 30               | 99.95% |                          |                           |
| Premium v4 P4mv4 | -                      | 16   | 128         | 250                 | 30               | 99.95% |                          |                           |
| Premium v4 P5mv4 | -                      | 32   | 256         | 250                 | 30               | 99.95% |                          |                           |
| Isolated v2 I1V2 | 195                    | 2    | 8           | 1000                | N/A              | 99.95% |                          |                           |

Select

\*ACU/vCPU is an approximation of the SKU's relative performance.

[Learn more about App Service pricing](#)

Applied filters to match the lab requirements.

azure.microsoft.com/en-us/pricing/details/app-service/linux/

All Bookmarks

Apply filters to customize pricing options to your needs.

Prices are estimates only and are not intended as actual price quotes. Actual pricing may vary depending on the type of agreement entered with Microsoft, date of purchase, and the currency exchange rate. Prices are calculated based on US dollars and converted using London closing spot rates that are captured in the two business days prior to the last business day of the previous month end. If the two business days prior to the end of the month fall on a bank holiday in major markets, the rate setting day is generally the day immediately preceding the two business days. This rate applies to all transactions during the upcoming month. Sign in to the [Azure pricing calculator](#) to see pricing based on your current program/offer with Microsoft. Contact an [Azure sales specialist](#) for more information on pricing or to request a price quote. See [frequently asked questions](#) about Azure pricing.

OS:

Linux

Region:

East US

Currency:

United States – Dollar (\$) USD

Display pricing by:

Month

**F1 (Free tier)** — 1 GB RAM, 60 min/day compute. Great for learning and development.

←→↺

azure.microsoft.com/en-us/pricing/details/app-service/linux/

☆🔖👤⋮

☐☐☐☐

All Bookmarks

OS:

Linux

Region:

East US

Currency:

United States – Dollar (\$) USD

Display pricing by:

Month

App Service Free Plan

The free plan is intended for trials, experimentation, and learning the service. There is no SLA for free plan and it is metered on a per app basis. Use of free plan for production workloads is not supported.

| Free Plan  | Cores                            | RAM  | Storage | Pay as you go |
|------------|----------------------------------|------|---------|---------------|
| F1<br>Free | Shared<br>(60 CPU minutes / day) | 1 GB | 1.00 GB | \$0           |

Dev/Test Pricing Available

Available for Visual Studio subscribers looking to run development and testing workloads, individually or as a team, the [dev/test offer](#) is billed at the same rate as Azure Hybrid Benefit pricing, with [savings of up to 55% on License Included prices](#). Dev/Test rates are now available on Linux App Service Premium v3.

Chat with Sales

B1 (Basic) — Always-on, custom domains at \$12.41/month.

←→↺

azure.microsoft.com/en-us/pricing/details/app-service/linux/

☆🔖👤⋮

☐☐☐☐

All Bookmarks

OS:

Linux

Region:

East US

Currency:

United States – Dollar (\$) USD

Display pricing by:

Month

based on the size and number of instances you run. Additional Azure savings such as savings plan and reserved instances are not available with this plan.

Existing customers looking for Standard plan, [click here](#).

| Basic Service Plan | Cores | RAM     | Storage | Pay as you go  |
|--------------------|-------|---------|---------|----------------|
| B1                 | 1     | 1.75 GB | 10 GB   | \$12.41/month  |
| B2                 | 2     | 3.50 GB | 10 GB   | \$24.82/month  |
| B3                 | 4     | 7 GB    | 10 GB   | \$48.910/month |

Chat with Sales

App Service Premium version 3 plan

Clicked to view Standard plans.

← → ↺

azure.microsoft.com/en-us/pricing/details/app-service/linux/

☆

🔖

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☰

All Bookmarks

OS:  
Linux

Region:  
East US

Currency:  
United States – Dollar (\$) USD

Display pricing by:  
Month

### App Service Basic plan

The Basic service plan is designed to run workloads that have low traffic requirements, and do not require advanced auto-scale and traffic management features. Pricing is based on the size and number of instances you run. Additional Azure savings such as savings plan and reserved instances are not available with this plan.

Existing customers looking for Standard plan, [click here](#).

| Basic Service Plan | Cores | RAM     | Storage | Pay as you go  |
|--------------------|-------|---------|---------|----------------|
| B1                 | 1     | 1.75 GB | 10 GB   | \$12.41/month  |
| B2                 | 2     | 3.50 GB | 10 GB   | \$24.82/month  |
| B3                 | 4     | 7 GB    | 10 GB   | \$48.910/month |

Chat with Sales

**S1 (Standard) — Auto-scaling, staging slots at \$69.35/month.**

← → ↺

azure.microsoft.com/en-us/pricing/details/app-service/linux-previous/

☆

🔖

👤

⋮

☰

All Bookmarks

OS:  
Linux Legacy

Region:  
East US

Currency:  
United States – Dollar (\$) USD

Display pricing by:  
Month

| Standard Service Plan | Cores | RAM     | Storage | Pay as you go  |
|-----------------------|-------|---------|---------|----------------|
| S1                    | 1     | 1.75 GB | 50 GB   | \$69.35/month  |
| S2                    | 2     | 3.50 GB | 50 GB   | \$138.70/month |
| S3                    | 4     | 7 GB    | 50 GB   | \$277.40/month |

**P1V2 (Premium Legacy) — Advanced scaling, VNet integration at \$73.73/month.**

← → ↺

azure.microsoft.com/en-us/pricing/details/app-service/linux-previous/

☆

All Bookmarks

OS:  
Linux Legacy

Region:  
East US

Currency:  
United States – Dollar (\$) USD

Display pricing by:  
Month

| Premium v2 Service Plan | Cores | RAM     | Storage | Pay as you go  |
|-------------------------|-------|---------|---------|----------------|
| P1v2                    | 1     | 3.50 GB | 250 GB  | \$73.73/month  |
| P2v2                    | 2     | 7 GB    | 250 GB  | \$147.46/month |
| P3v2                    | 4     | 14 GB   | 250 GB  | \$294.19/month |

Isolated v1 service plan

App Service Environment version 1 and version 2, that use the Isolated v1 service plan, will be [retired](#) on August 31, 2024. As of January 29, 2024, you can no longer create new App Service Environment v1 and v2 resources using any of the available methods including ARM/Bicep templates, Azure Portal, Azure CLI, or REST API. [Chat with Sales](#)

Cancelled App Service creation — exploration complete.

← → ↺

portal.azure.com/#view/WebsitesExtension/AppServiceWebAppCreateV3Blade

⌵ 🔍 ☆

All Bookmarks

Microsoft Azure

Search resources, services, and docs (G+I)

Copilot

oluwaseun.beicks@gmail...  
DEFAULT DIRECTORY (OLUWASEUN...)

Home > App Services

Create Web App

Region: East US

Not finding your App Service Plan? Try a different region or select your App Service Environment.

Pricing plans

App Service plan pricing tier determines the location, features, cost and compute resources associated with your app. [Learn more](#)

Linux Plan (East US) \* ⓘ

(New) ASP-rgcloudfirstdev-a7f0  
[Create new](#)

Pricing plan

Free F1 (Shared infrastructure)  
[Explore pricing plans](#)

Zone redundancy

An App Service plan can be deployed as a zone redundant service in the regions that support it. Your initial instance count will be set based on your zone redundancy configuration. To ensure you'll be able to enable zone redundancy at any point in the lifecycle of your app, enable zone redundancy now. You can decrease the instance count after the App Service plan is created. [Learn more](#)

Zone redundancy

☐ Enabled: Your App Service plan and the apps in it will be zone redundant. The minimum App Service plan instance count will be two.

☒ Disabled: Your App Service plan and the apps in it will not be zone redundant. The minimum App Service plan instance count will be one.

Review + create

< Previous

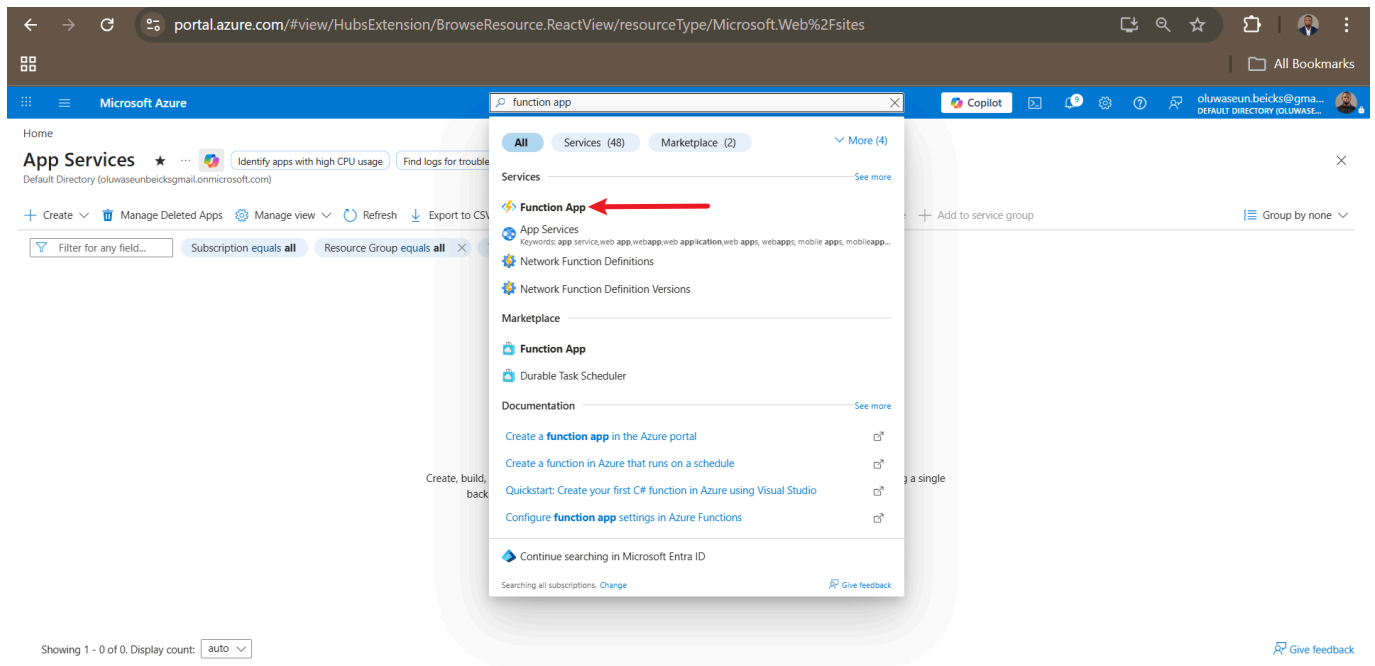
Next: Database >



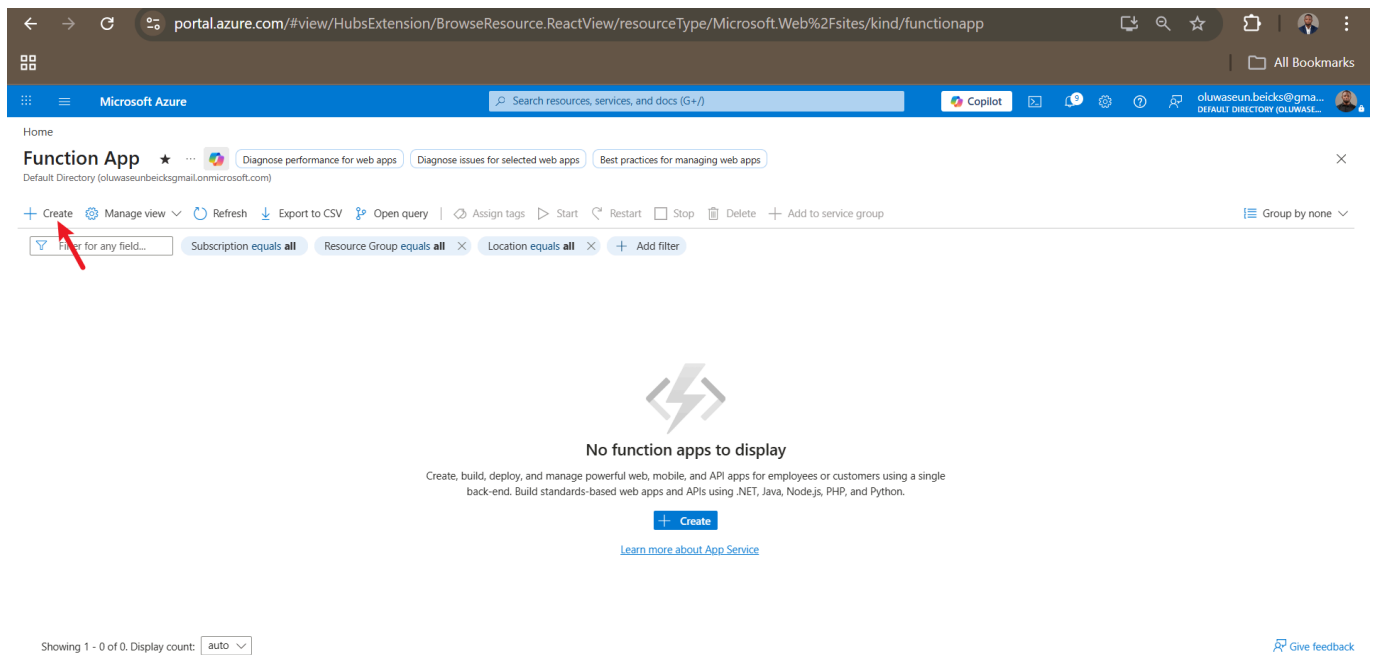
Azure Functions (Serverless) — Explore Configuration

Searched for **Function App**.

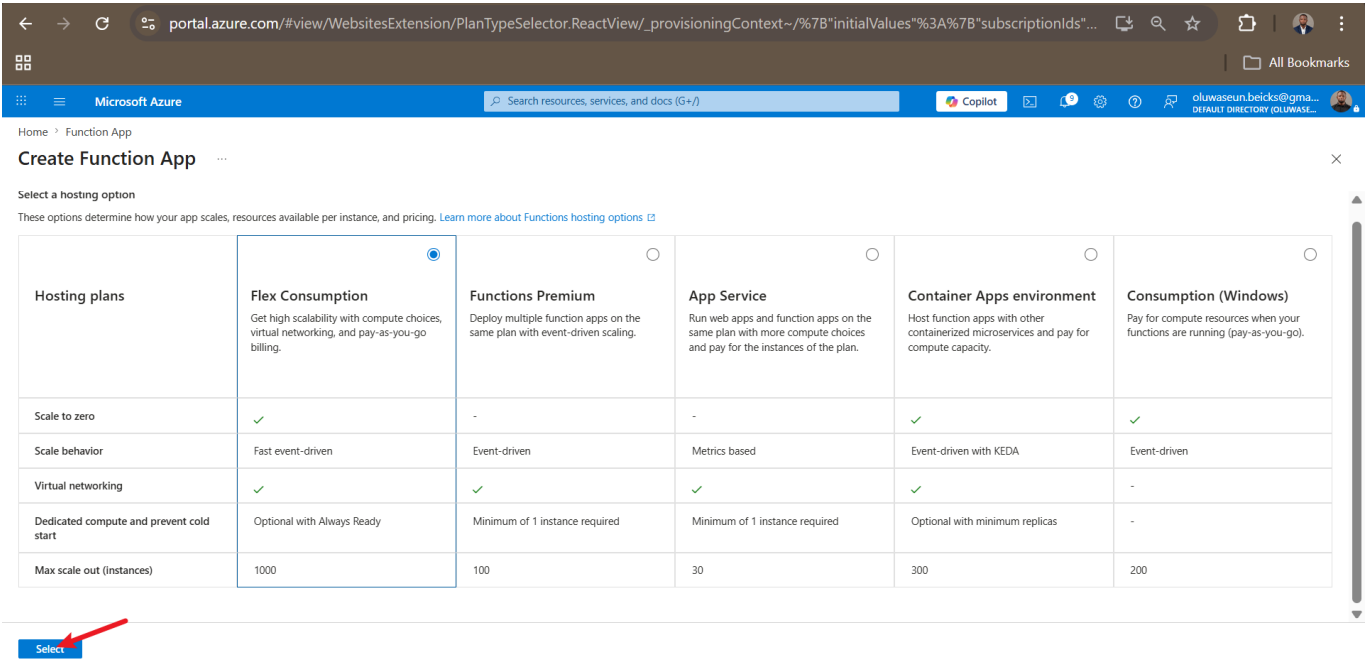




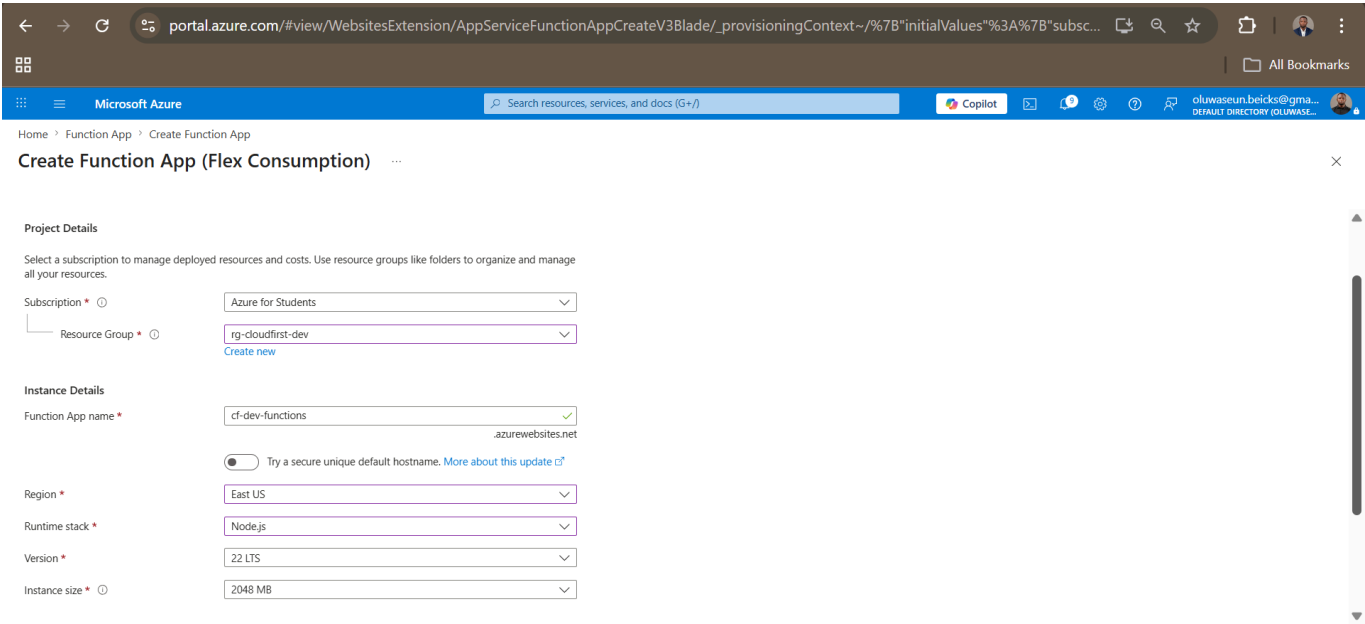
Clicked **Create**.



Selected **Flex Consumption** (pay-as-you-go serverless hosting) to demonstrate the event-driven, per-execution billing model — the first 1 million executions per month are free.

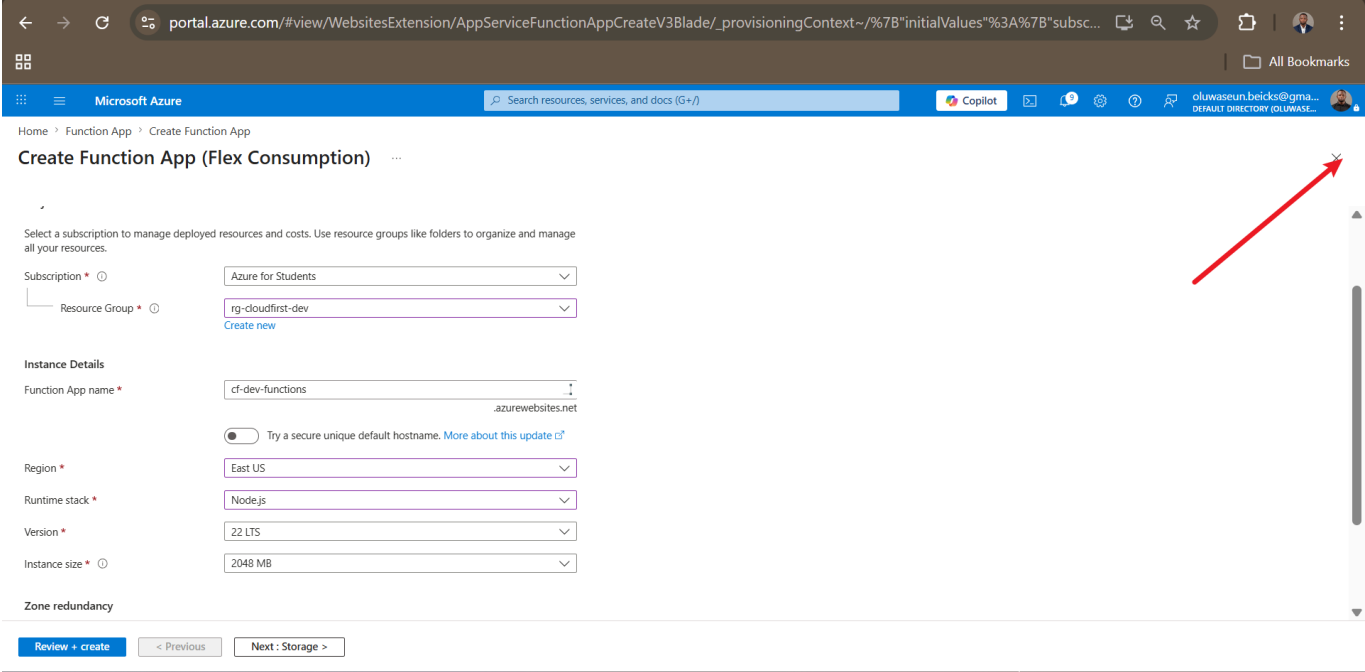


Configured: Subscription · **rg-cloudfirst-dev** · Name: **cf-dev-functions** · Runtime: **Node 22 LTS** · OS: **Linux** · Region: **East US**.



**Key Observation vs. App Service:** No infrastructure management at all. You provide only the code and runtime. Azure handles scaling, OS, patching, and availability automatically.

Cancelled creation — exploration complete.

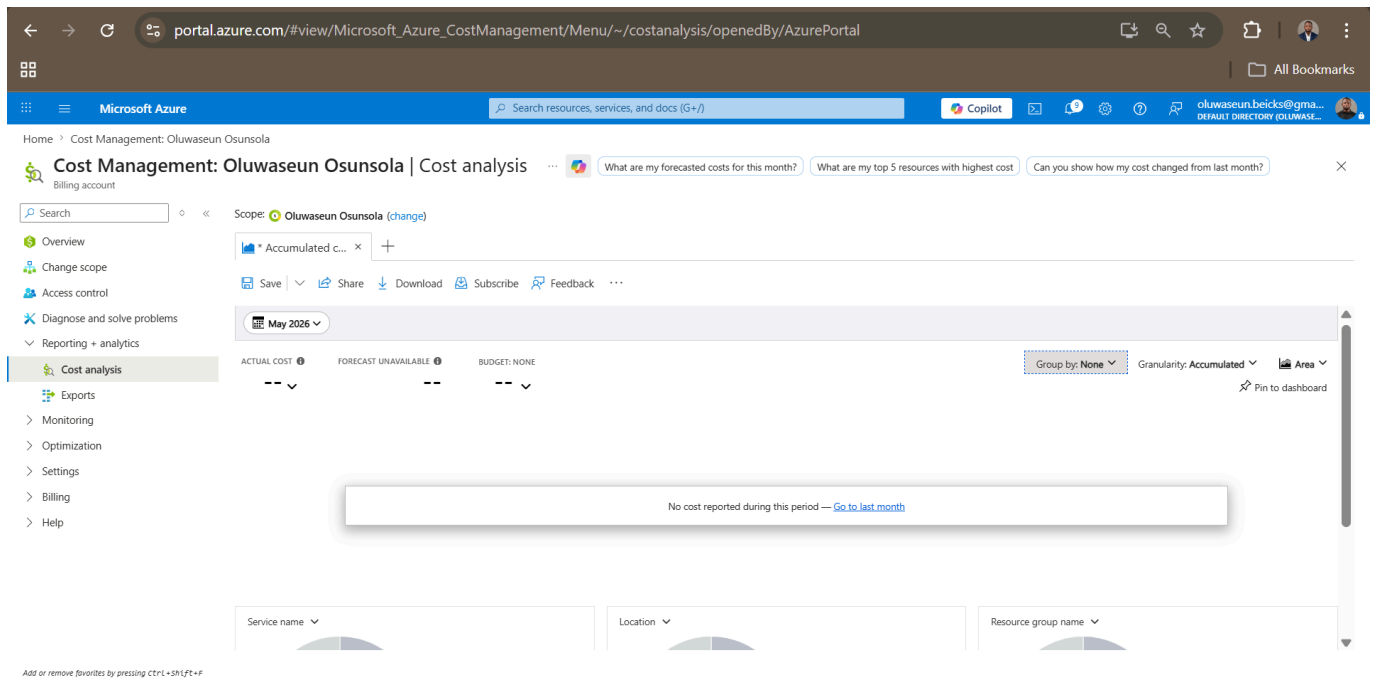


Compute Service Decision Matrix

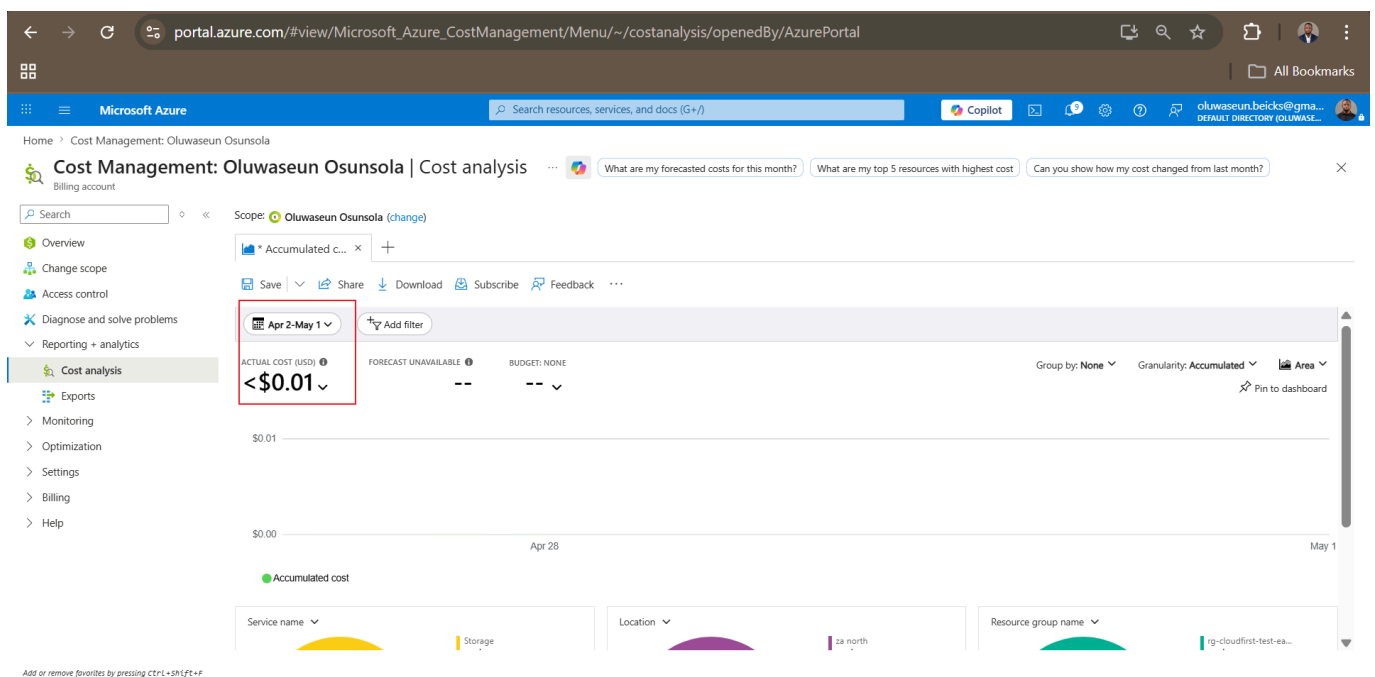
| Requirement                       | Best Service                 | Why                                   |
|-----------------------------------|------------------------------|---------------------------------------|
| Full OS control / legacy app      | Virtual Machine (IaaS)       | Complete OS, driver, registry control |
| Lift-and-shift migration          | Virtual Machine (IaaS)       | Minimal architectural changes         |
| Web app or REST API               | App Service (PaaS)           | Built-in scaling, managed runtime     |
| Minimal infrastructure management | App Service (PaaS)           | Azure manages OS, patching, runtime   |
| Code runs only when triggered     | Azure Functions (Serverless) | Pay per execution, not per hour       |
| Event-driven / queue processing   | Azure Functions (Serverless) | Purpose-built for event architecture  |
| Container orchestration           | AKS                          | Managed containerized workloads       |

Step 6 — Review Cost Management Interface

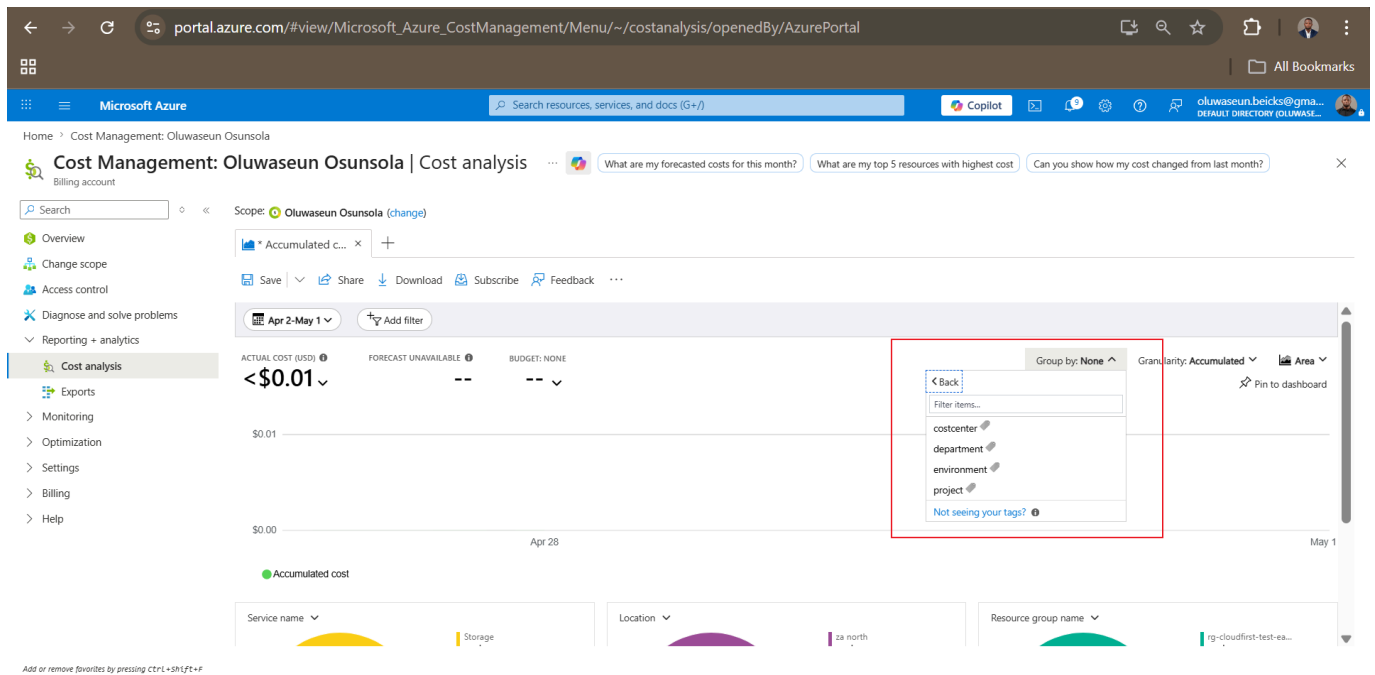
Searched for **Cost Management + Billing**, clicked into **Reports + Analytics** → **Cost Analysis**. No costs accumulated in the current month — expected for a newly created environment.



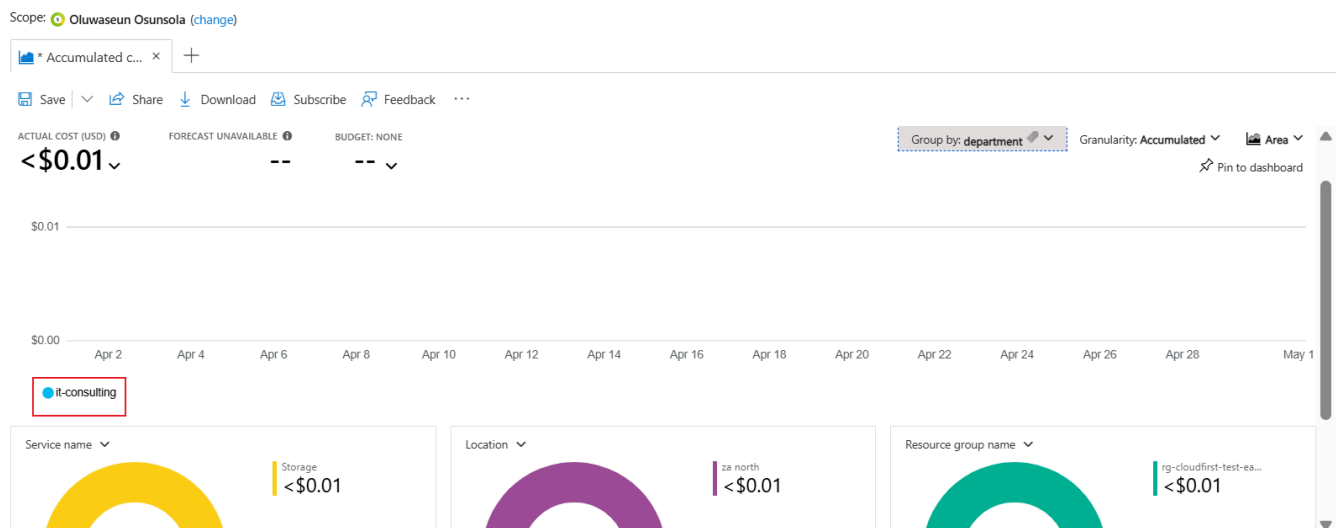
Filtered to **Last 30 days** and observed **\$0.01 USD** accumulated — a small charge from a previously deployed Storage Account in the **rg-cloudfirst-test-eastus** Resource Group from an earlier lab.



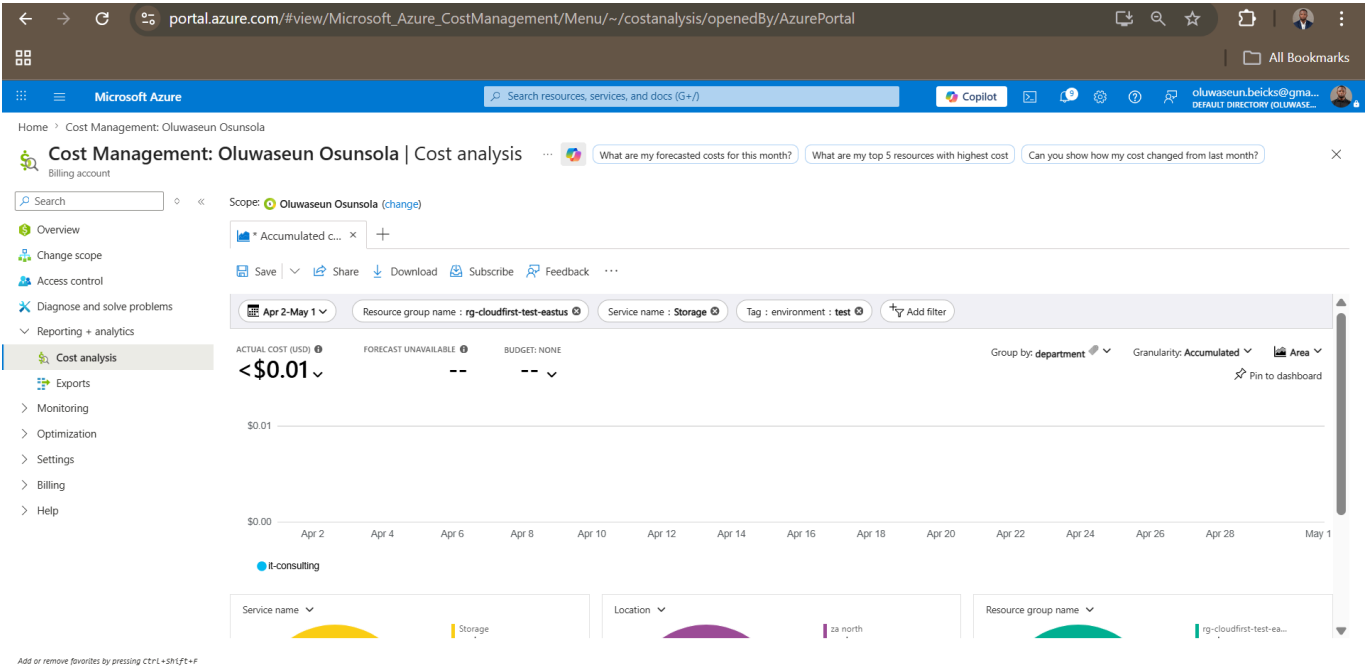
Applied **Group by: Tag** to break down the cost by department tag.



Applied the **Department** tag filter — identified that the **IT-Consulting** department incurred the \$0.01 charge, traced to a Storage Account resource.



Applied combined filters — **Resource Group:** **rg-cloudfirst-test-eastus** · **Service name:** Storage · **Environment:** Test — to produce a fully scoped, filtered cost view.



**Key Insight:** This is tag-based cost attribution in action. The **Department**, **Environment**, and **Project** tags applied in the earlier lab enabled this precise cost drill-down — demonstrating exactly why consistent tagging across all resources is non-negotiable from day one.



## Concepts Demonstrated at a Glance

| Concept                              | What Was Demonstrated                                                           |
|--------------------------------------|---------------------------------------------------------------------------------|
| VM Sizing Families                   | B (Burstable), D (General), E (Memory) compared hands-on with real pricing      |
| Savings Plans vs. Reserved Instances | 1yr/3yr options compared — up to 60% savings over pay-as-you-go                 |
| Tag Non-Inheritance                  | Tags manually reapplied to VM because RG tags do not propagate                  |
| ARM Template / IaC                   | Full VM config exported as downloadable JSON for automation reuse               |
| Auto-generated VM Resources          | VNet, Subnet, Public IP, NIC, NSG all created automatically                     |
| IaaS vs. PaaS vs. Serverless         | Configuration complexity decreases as management responsibility shifts to Azure |
| App Service Pricing Tiers            | F1 (Free) → B1 (\$12.41) → S1 (\$69.35) → P1V2 (\$73.73) compared               |
| Serverless Billing Model             | Azure Functions Flex Consumption — pay per execution, not per hour              |
| Azure Advisor 5 Pillars              | Cost, Security, Reliability, Operational Excellence, Performance reviewed       |
| Tag-Based Cost Attribution           | Department tag identified which team incurred charges on which resource         |



## Key Takeaways

**Right-sizing starts before deployment.** The Azure Pricing Calculator is the primary tool for budget planning, service comparison, and stakeholder communication. Use it before committing any infrastructure.

**Tags must be applied explicitly — they don't inherit.** RBAC permissions, Azure Policy assignments, and Resource Locks cascade from Resource Group to resource. Tags do not. Enforce tagging at scale using Azure Policy.

**The compute spectrum: control vs. convenience.** VMs (IaaS) give maximum control with maximum management responsibility. App Service (PaaS) reduces overhead to application code. Azure Functions (Serverless) removes infrastructure entirely — you manage only the code.

**ARM templates capture every portal decision as IaC.** The "Download template for automation" button on the Review + Create screen exports a JSON ARM template ready for version control, parameterisation, and pipeline integration.

**Reserved Instances provide up to 60% savings for predictable workloads.** Pay-as-you-go is right for variable or short-lived workloads. For 24/7 production workloads with predictable demand, 1-year or 3-year Reserved Instances are the financially responsible choice.

**Cost Management needs tags to tell the story.** Without consistent tagging, Cost Analysis shows total spend but cannot attribute it. With tags, it becomes a full financial reporting tool — filterable by department, environment, project, and resource type.



## AZ-900 Exam Quick Reference

### VM Series:

- **B** = Burstable → Dev/test, variable workloads
- **D** = Daily Use → General purpose production
- **E** = Enormous Memory → Databases, caching
- **F** = Fast CPU → Batch, analytics
- **M** = Massive Memory → Terabyte-scale

### SLA by VM Configuration:

| Configuration                 | SLA    |
|-------------------------------|--------|
| Single VM + Standard SSD      | No SLA |
| Single VM + Premium SSD       | 99.9%  |
| 2+ VMs in Availability Set    | 99.95% |
| VMs across Availability Zones | 99.99% |

### Cost Tools — Know the Difference:

| Tool                      | Purpose                                      |
|---------------------------|----------------------------------------------|
| Pricing Calculator        | Estimate costs <b>before</b> deployment      |
| Cost Management + Billing | Monitor <b>actual</b> spending and forecasts |

| Tool           | Purpose                                        |
|----------------|------------------------------------------------|
| Azure Advisor  | AI-powered <b>optimization recommendations</b> |
| TCO Calculator | Compare on-premises vs. Azure migration costs  |

#### What Inherits from Resource Groups:

-  RBAC permissions
-  Azure Policy assignments
-  Resource Locks
-  Tags (must be applied individually)

---

## Resources

- [Azure Portal](#)
- [Azure Pricing Calculator](#)
- [Azure Virtual Machine Sizes](#)
- [App Service Pricing](#)
- [Azure Functions Pricing](#)
- [Azure Advisor Documentation](#)
- [Azure Cost Management](#)

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*Lab completed by Oluwaseun Osunsola [[Linkedln](#)] as part of the Microsoft Azure Fundamentals (AZ-900) certification preparation — Section 4: Azure Compute Options.*